

EVENTFORCE MANAGEMENT SYSTEM- SALESFORCE IMPLEMENTATION

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EVENTFORCE MANAGEMENT SYSTEM SALESFORCE MANAGEMENT

Abstract:

The EventForce Management System is a Salesforce-based Customer Relationship Management (CRM) solution developed for Event Planner Solutions. The primary purpose of this project is to provide a centralized platform for managing the complete event-planning lifecycle, including client management, event scheduling, venue management, vendor coordination, cancellation approvals, customer feedback, automated reminders, reporting, and operational monitoring.

Traditional event-management processes often involve multiple manual activities such as maintaining client details, checking venue availability, coordinating vendors, following up with clients, and processing cancellations. These activities, when handled separately, can result in duplicate data, scheduling conflicts, communication delays, and difficulty in monitoring event operations.

To address these challenges, the EventForce Management System uses Salesforce's declarative and programmatic capabilities. The system contains six custom objects: **Event, Client, Vendor, Venue, Feedback, and Event Vendor**. Salesforce features such as custom objects, fields, relationships, validation rules, lookup filters, Flow automation, Approval Processes, email notifications, Apex Triggers, Apex Classes, Batch Apex, Scheduled Apex, Profiles, Roles, Permission Sets, Organization-Wide Defaults, Sharing Rules, Reports, and Dashboards have been implemented.

The system includes automated venue-status management that marks venues as **Reserved** when events are confirmed and **Available** when events are cancelled. It also includes double-booking prevention logic that blocks conflicting venue reservations. A three-day client reminder Flow automates communication before confirmed events. An approval process controls event cancellations with manager-based routing and email notifications. Batch Apex and Scheduled Apex automatically mark past events as **Completed**.

Security and access control are implemented through Profiles, Roles, Permission Sets, Organization-Wide Defaults, and Sharing Rules. Reports and dashboards provide management visibility into upcoming events and operational performance.

The resulting application provides a centralized, controlled, and automated environment for managing event operations. The project demonstrates practical implementation of Salesforce administration, declarative automation, programmatic development, security configuration, reporting, and testing.

Keywords: Salesforce CRM, Event Management, Custom Objects, Flow Automation, Approval Process, Apex Programming, Batch Apex, Security Configuration, Reports, Dashboards, Lightning App

1. INTRODUCTION

1.1 Project Overview

The EventForce Management System is a Salesforce-based CRM solution developed to centralize and streamline event planning and management operations. The system manages clients, events, venues, vendors, feedback, and event cancellations within a structured Salesforce platform.

The solution uses custom objects, relationships, validation rules, Flows, Approval Processes, Apex Triggers and Classes, Reports, and Dashboards to automate key business processes. It supports venue availability management, prevention of double bookings, automated event reminders, cancellation approvals, feedback management, and operational reporting.

The project aims to reduce manual effort, improve booking coordination and data accuracy, enhance communication with clients and operational teams, and provide better visibility into event activities through a centralized Salesforce platform.

The EventForce Management System was developed as a four-member team project with the objective of creating a comprehensive CRM solution for Event Planner Solutions. The implementation covers the complete lifecycle from event creation to event completion,

demonstrating how Salesforce can transform a manual event-management process into a centralized and automated CRM solution.

Key System Characteristics:

- **Platform:** Salesforce Developer Edition (Lightning Experience)
- **Architecture:** Multi-layer (Data, Automation, Security, Presentation)
- **Objects:** Six custom objects with defined relationships
- **Automation:** Record-triggered Flows, Approval Processes, Email Alerts
- **Programming:** Apex Classes, Triggers, Batch Apex, Scheduled Apex
- **Security:** Profiles, Roles, Permission Sets, OWD, Sharing Rules
- **Analytics:** Reports and Dashboards

The Event Planner application serves as the central workspace for all event-management activities. The navigation menu provides direct access to Events, Clients, Vendors, Venues, Feedbacks, Reports, and Dashboards. Users can create, view, and manage records from this unified interface.

The application is designed to support different user roles including Event Administrators, Event Coordinators, Vendor Managers, and Clients. Each role has appropriate access permissions configured through Salesforce's security model.

The system architecture follows Salesforce best practices for enterprise application development, separating concerns into data management, business logic, security, and presentation layers.

1.2 Purpose

The purpose of the EventForce Management System is to provide a centralized Salesforce platform for managing and automating event planning operations. The system is designed to:

Primary Objectives:

1. **Centralize Information:** Consolidate client, event, venue, vendor, and feedback data in a single platform
2. **Streamline Booking:** Simplify event booking and venue reservation processes
3. **Prevent Conflicts:** Eliminate double booking of venues through automated Apex logic

4. **Automate Communication:** Implement automated event reminders and cancellation approval processes
5. **Improve Coordination:** Enhance collaboration between clients, event coordinators, vendors, and venue managers
6. **Provide Visibility:** Deliver reports and dashboards for operational monitoring and decision-making
7. **Reduce Manual Effort:** Minimize repetitive administrative tasks through automation
8. **Ensure Data Quality:** Maintain accurate and consistent data through validation rules
9. **Control Access:** Implement role-based security to protect sensitive information
10. **Enable Scalability:** Build a foundation that can grow with business requirements

Business Value Proposition:

The EventForce Management System delivers significant business value by transforming manual, fragmented event-management processes into a centralized, automated, and secure CRM solution. The system reduces the risk of scheduling conflicts, improves client communication, and provides management with real-time operational visibility.

1.3 Scope of the Project

The scope of the EventForce Management System defines the boundaries of the implementation and identifies what is included and excluded from the current phase.

In Scope:

Category	Components
Data Management	Client management, Event management, Venue management, Vendor management, Feedback management
Relationships	Lookup relationships, Master-Detail relationships, Many-to-Many through junction object
Automation	Validation rules, Formula fields, Lookup filters, Record-triggered Flows
Approval	Event cancellation approval process with manager routing

Category	Components
Notifications	Email templates for cancellation notifications
Reminders	Three-day client reminder automation
Programming	Apex Classes, Apex Triggers, Batch Apex, Scheduled Apex
Security	Profiles, Roles, Permission Sets, OWD, Sharing Rules
Analytics	Reports, Dashboards
UI	Lightning Application with navigation
Data Migration	Data Import Wizard for sample data loading

Out of Scope (Future Enhancements):

Category	Components
Payment	Online payment processing
Integration	External calendar integration (Google Calendar, Outlook)
Messaging	SMS or WhatsApp notification integration
Portal	Customer self-service portal
Mobile	Dedicated mobile application
AI	AI-based event recommendations
Production	Production deployment and migration
Advanced Analytics	CRM Analytics / Einstein Analytics

Assumptions:

1. Users have basic familiarity with Salesforce Lightning interface
2. The organization has Salesforce licenses for required users
3. Email deliverability is configured for notifications
4. Sample data is representative of actual business scenarios
5. The Developer Edition org is used for demonstration purposes

Constraints:

1. Salesforce Developer Edition has storage and API limits
2. Only two Salesforce licenses available in Developer Edition
3. Email sending limits apply in Developer Edition
4. Scheduled jobs may be delayed based on system load
5. Batch Apex has governor limits on records processed

1.4 Organization of the Report

This report is organized into eleven sections:

Section 1: Introduction – Provides an overview of the project, its purpose, scope, and organization of the report.

Section 2: Ideation Phase – Documents the problem statement, empathy map, brainstorming, and idea prioritization that led to the selection of Salesforce CRM as the solution platform.

Section 3: Requirement Analysis Phase – Presents the customer journey map, functional and non-functional requirements, data flow diagrams, and technology stack.

Section 4: Project Design Phase – Describes the problem-solution fit, proposed solution, solution architecture, and data model.

Section 5: Project Planning & Scheduling – Details the Agile methodology, product backlog, sprint schedule, and velocity calculations.

Section 6: Project Development Phase – Provides comprehensive documentation of all implementation milestones from account creation through security configuration.

Section 7: Functional & Performance Testing – Documents the testing approach, test cases, and results.

Section 8: Advantages & Disadvantages – Discusses the benefits and limitations of the implemented solution.

Section 9: Conclusion – Summarizes the project outcomes and learnings.

Section 10: Future Scope – Identifies potential enhancements and extensions.

Section 11: Team Contribution – Documents the contribution of each team member.

2. IDEATION PHASE

2.1 Problem Statement

The ideation phase began with identifying the key problems faced by Event Planner Solutions in managing event operations. A structured approach was used to document customer problems using the standard Problem Statement Template.

Problem Statement Template:

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Event Operations Manager	manage event planning operations efficiently	client, event, venue, and vendor information is difficult to manage centrally	event management activities require coordination across multiple areas	frustrated and concerned
PS-2	Event Coordinator	create and manage event bookings efficiently	event details, client information, and venue reservations require multiple manual activities	there is no centralized platform for managing the complete event lifecycle	overwhelmed and frustrated
PS-3	Venue Manager	manage venue availability accurately	the same venue may be assigned to multiple events	venue bookings need proper availability tracking and double-booking prevention	concerned and uncertain

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-4	Vendor Manager	coordinate vendors for different events	vendor details and event assignments need to be managed separately	events can require multiple vendors and services	overwhelmed and concerned
PS-5	Event Coordinator	communicate event information and reminders on time	reminders have to be handled manually	there is a need to notify clients before upcoming events	stressed and concerned
PS-6	Event Administrator	manage event cancellation requests systematically	cancellation requests require approval and status updates	cancellation decisions need a controlled approval process	frustrated and uncertain
PS-7	Event Coordinator	collect and manage client feedback	feedback information needs to be maintained along with event and client details	feedback is related to both events and clients	concerned about customer satisfaction
PS-8	Event Operations Manager	monitor event operations and performance	operational information is not easily visible in one place	event data needs to be organized into reports and dashboards	concerned and less confident in decision-making

Table 1: Problem Statement Template

Analysis of Problems:

The problem statements reveal several recurring themes:

1. **Fragmentation:** Information is scattered across multiple systems and formats
2. **Manual Processes:** Many activities require manual intervention
3. **Lack of Visibility:** Management cannot easily monitor operations

4. **Coordination Challenges:** Multiple stakeholders need to collaborate
5. **Data Integrity:** Risk of errors and inconsistencies
6. **Communication Gaps:** Clients may not receive timely updates
7. **Control Requirements:** Cancellations need approval workflows
8. **Security Concerns:** Different users need different access levels

Impact Assessment:

Problem Area	Business Impact	Severity
Fragmented Information	Difficulty in decision-making, data inconsistencies	High
Manual Booking Processes	Time wastage, human errors, delays	High
Double Booking Risk	Client dissatisfaction, revenue loss, reputation damage	Critical
Vendor Coordination	Service delays, quality issues	Medium
Manual Reminders	Missed communications, client dissatisfaction	Medium
Uncontrolled Cancellations	Revenue loss, process confusion	High
Feedback Management	Inability to improve services	Medium
Limited Visibility	Poor decision-making, missed opportunities	High
Security Gaps	Data breaches, unauthorized access	Critical



2.2 Empathy Map Canvas

The Empathy Map Canvas helps understand the Event Coordinator's perspective – what they think, feel, see, hear, say, and do. This understanding informs the design of the EventForce Management System.

User Persona: Event Coordinator

Role: Event Coordinator

Primary Responsibility: Manage event bookings, client requirements, venue reservations, vendor coordination, reminders, and event status updates.



Empathy Map:

Quadrant	Observations
Think & Feel	Wants to manage events smoothly, avoid booking conflicts, and provide a reliable experience to clients. Feels responsible for coordinating multiple stakeholders. Hopes everything goes as planned. Feels stressed when there are last-minute changes. Worries about missing important details.
Hear	Client requirements and expectations. Updates and confirmations from vendors. Venue availability and constraints from venue managers. Guidance and approvals from event administrators. Feedback and concerns from clients. Suggestions and queries from the event team. Complaints about double bookings or delays.
See	Event details, client information, and bookings. Venue availability and schedules. Vendor assignments and service details. Cancellation requests and approval status. Client feedback and event history. Reports and dashboards on event performance. Multiple spreadsheets and email threads.

Quadrant	Observations
Say & Do	Communicates with clients, vendors, and venue managers. Creates and updates event records. Coordinates venue reservations and vendor assignments. Follows up on client requirements and approvals. Sends reminders and ensures on-time updates. Monitors event progress and resolves issues. Juggles multiple tasks simultaneously.
Pain	Manual coordination and multiple communication channels. Fragmented information across different systems. Risk of venue double bookings. Difficulty in tracking multiple event activities. Time-consuming approval and cancellation processes. Stress due to last-minute changes. Lack of centralized information. Difficulty generating reports.
Gain	Centralized and easy access to event information. Automated reminders and approval processes. Accurate venue availability and no double bookings. Better coordination with clients, vendors, and venue managers. Improved operational visibility through reports and dashboards. Successful events and higher client satisfaction. Reduced stress and more efficient workflows.

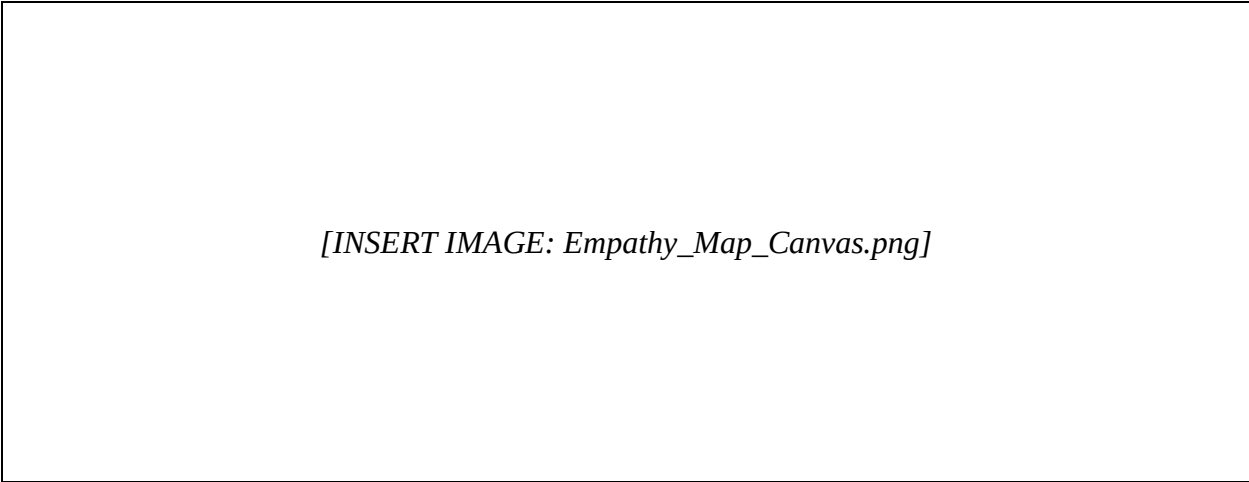


Figure 2: Empathy Map Canvas – Event Coordinator

Key Insights from Empathy Map:

1. **Centralization Need:** The Event Coordinator desperately needs a single source of truth for all event-related information
2. **Automation Desire:** Manual reminders, approvals, and status updates consume valuable time

- 3. **Conflict Prevention:** Double booking is a major source of stress and client dissatisfaction
- 4. **Visibility Requirement:** Reports and dashboards would help track multiple events simultaneously
- 5. **Communication Efficiency:** Multiple channels (phone, email, messaging) create confusion and missed updates
- 6. **Process Control:** Cancellation approvals need structure to prevent unauthorized changes

2.3 Brainstorming & Idea Prioritization

The team conducted brainstorming sessions to identify potential solutions and evaluate them against business requirements.

Ideas Considered:

Idea	Approach	Key Limitation
Manual Registers	Maintain event, client, venue, and vendor details manually in physical registers	Time-consuming, difficult to maintain, no automation, error-prone
Spreadsheet Tracking	Track bookings and event information using Microsoft Excel or Google Sheets	Limited automation, no real-time coordination, version control issues, no security
Standalone Portal	Develop a dedicated web portal for event management using custom development	Higher development cost, longer timeline, maintenance overhead, requires hosting
Salesforce CRM	Centralize event operations on Salesforce platform using declarative and programmatic features	Requires proper configuration and implementation, licensing cost

Table 2: Idea Prioritization Matrix

Evaluation Criteria:

Criteria	Weight	Manual Registers	Spreadsheet	Standalone Portal	Salesforce CRM
Automation	25%	1	2	4	5
Scalability	20%	1	2	4	5
Real-time Visibility	20%	1	2	4	5

Criteria	Weight	Manual Registers	Spreadsheet	Standalone Portal	Salesforce CRM
Security	15%	1	2	4	5
Development Effort	10%	5	4	2	3
Maintenance	10%	2	3	2	4
Weighted Score	100%	1.45	2.15	3.60	4.75



Final Selection:

Salesforce CRM was selected due to:

1. **Automation** – Automates reminders, approvals, status updates, and business processes through Flows, Approval Processes, and Apex
2. **Scalability** – Supports expanding event operations, users, records, and business requirements without significant rework
3. **Real-Time Visibility** – Provides centralized data, reports, and dashboards for operational monitoring
4. **Security** – Built-in security model with Profiles, Roles, Permission Sets, OWD, and Sharing Rules
5. **Integration Capability** – Can be integrated with external systems through APIs
6. **Mobile Access** – Salesforce mobile app provides access from anywhere

7. **Proven Platform** – Enterprise-grade platform with extensive documentation and community support

Decision: Salesforce CRM provides the best fit for creating a centralized, automated, and scalable EventForce Management System.

2.4 Decision Rationale

The decision to use Salesforce CRM was based on a thorough analysis of alternatives and alignment with business requirements.

Why Salesforce CRM:

Requirement	Salesforce Capability
Centralized data management	Custom objects with relationships
Automated reminders	Record-triggered Flows with scheduled paths
Controlled cancellations	Approval Processes with email alerts
Venue availability tracking	Apex Triggers and Classes
Double booking prevention	Apex Triggers with validation logic
Automatic event completion	Batch Apex and Scheduled Apex
Role-based security	Profiles, Roles, Permission Sets, OWD, Sharing Rules
Operational visibility	Reports and Dashboards
User-friendly interface	Lightning App with navigation
Data import capability	Data Import Wizard

Cost-Benefit Analysis:

Factor	Consideration
Initial Investment	Developer Edition available free; production requires licensing
Development Time	Declarative tools reduce development time significantly
Maintenance	Salesforce handles platform maintenance and upgrades
Scalability	Additional users, objects, and automation can be added incrementally
Training	Salesforce skills are valuable and widely applicable

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

The Customer Journey Map visualizes the end-to-end experience of a client interacting with Event Planner Solutions through the EventForce Management System.

Stage	Customer Journey Stage	What Happens	Customer Emotion
1	Event Enquiry & Client Registration	Client requirements and contact details are captured in Salesforce.	Hopeful, expectant
2	Event Planning & Booking	Event details such as event type, date, budget, and status are created and managed.	Excited, involved
3	Venue Selection & Reservation	Suitable venues are identified, availability is checked, and reservations are managed while preventing double bookings.	Relieved, confident
4	Vendor Coordination	Vendors and required services are assigned to events and their availability/status is coordinated.	Assured, informed
5	Event Confirmation & Communication	Confirmed events trigger automated communication, including reminders sent to clients before the event.	Reassured, prepared
6	Event Execution & Monitoring	Event progress and operational information are monitored through Salesforce records, reports, and dashboards.	Engaged, satisfied
7	Cancellation & Approval	Cancellation requests are submitted through the approval process, with appropriate status updates and notifications.	Informed, respected
8	Feedback & Closure	Client feedback is captured after the event, and completed events are updated for operational tracking and future analysis.	Valued, appreciated

Table 3: Customer Journey Map



Figure 3: Customer Journey Map – From Enquiry to Everlasting Memories

Stage Details:

Stage 1: Event Enquiry & Client Registration

- Client contacts Event Planner Solutions with event requirements
- Event Coordinator creates Client record in Salesforce
- Details captured: Name, Email, Phone, Address, Country, City
- Email validation ensures accurate contact information

Stage 2: Event Planning & Booking

- Event Coordinator creates Event record
- Details captured: Event Name, Type, Date, Status
- Event Budget automatically calculated based on Event Type
- Event linked to Client through lookup relationship

Stage 3: Venue Selection & Reservation

- Event Coordinator selects venue from available options
- Venue availability checked automatically
- Double-booking prevention trigger validates venue-date combination
- Venue status updated to Reserved upon confirmation

Stage 4: Vendor Coordination

- Event Coordinator associates vendors through EventVendor junction object
- Multiple vendors can be assigned to a single event
- Vendor details include Service Type, Contact Information, Status

Stage 5: Event Confirmation & Communication

- Event Status changed to Confirmed
- Three-day reminder Flow activated
- Client receives automated reminder email before event
- Event Coordinator receives confirmation

Stage 6: Event Execution & Monitoring

- Event progress tracked through Salesforce records
- Reports provide visibility into upcoming events
- Dashboard shows operational metrics

Stage 7: Cancellation & Approval

- If cancellation required, status changed to Pending Cancellation
- Approval process initiated automatically
- Manager reviews and approves/rejects
- Client notified of cancellation decision

Stage 8: Feedback & Closure

- Client provides feedback after event
- Feedback record created with Rating and Comments
- Feedback linked to Event and Client
- Past events automatically marked Completed by Batch Apex

3.2 Solution Requirements

3.2.1 Functional Requirements

Functional requirements define what the system should do – the specific behaviors and functions.

FR No	Requirement	Description	Priority
FR-1	Event Management	Users should be able to create and manage event records with details including name, type, date, status, client, venue, and budget	High
FR-2	Client Management	Users should be able to create and maintain client information including name, email, phone, address, country, and city	High
FR-3	Venue Management	Users should be able to maintain venue information including name, address, capacity, and availability status	High
FR-4	Vendor Management	Users should be able to manage vendor information including name, email, phone, service type, and status	High
FR-5	Event-Vendor Assignment	System should allow multiple vendors to be associated with events through a junction object	High
FR-6	Venue Availability & Double-Booking Prevention	System should track venue availability and prevent double booking through Apex logic	Critical
FR-7	Automated Event Reminders	System should automatically send reminder emails to clients three days before confirmed events	High
FR-8	Event Cancellation Approval	Cancellation requests should follow an approval process with manager review	High
FR-9	Feedback Management	Users should be able to record customer feedback with rating and comments	Medium
FR-10	Reports & Dashboards	Management should be able to view upcoming events and operational metrics through reports and dashboards	High
FR-11	Event Status Automation	System should automatically update venue status based on event status changes	High

FR No	Requirement	Description	Priority
FR-12	Data Migration	System should support data import through Data Import Wizard	Medium
FR-13	Email Validation	System should validate client email addresses before saving records	Medium
FR-14	Feedback Lookup Filter	System should restrict event selection based on selected client when creating feedback	Medium
FR-15	Automatic Event Completion	System should automatically mark past events as Completed	Medium

Table 4: Functional Requirements

3.2.2 Non-Functional Requirements

Non-functional requirements define how the system should perform – quality attributes and constraints.

NFR	Category	Description	Target
NFR-1	Usability	Lightning App for easy event management and navigation	Intuitive interface requiring minimal training
NFR-2	Security	Profile-, role-, and permission-based access control	Users see only data relevant to their role
NFR-3	Reliability	Accurate automation, approvals, and venue booking validation	99.9% automation success rate
NFR-4	Performance	Efficient processing of event and operational data	Sub-second response for standard operations
NFR-5	Availability	Cloud-based Salesforce access for authorized users	99.9% platform availability
NFR-6	Scalability	Supports growing events, clients, vendors, venues, and users	Handle 10x current volume without redesign

NFR	Category	Description	Target
NFR-7	Maintainability	Flows, Apex, validation rules, and security can be monitored and updated	Modular design for easy updates
NFR-8	Data Integrity	Validation rules, lookup filters, and relationships maintain consistency	Zero orphaned records
NFR-9	Auditability	Track field history and approval history	Complete audit trail
NFR-10	Portability	Solution can be deployed to other Salesforce orgs	Metadata-based deployment

Table 5: Non-Functional Requirements

3.3 Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how data moves through the EventForce Management System.

Level 0 DFD (Context Diagram):

Client	EventForce Management System	Vendor
Manager	Reports & Dashboards	Venue Manager

Level 1 DFD (Main Processes):

Client Management	Event Management	Venue Management
Vendor Management	Event Vendor Assignment	Feedback Management
	Approval Process	

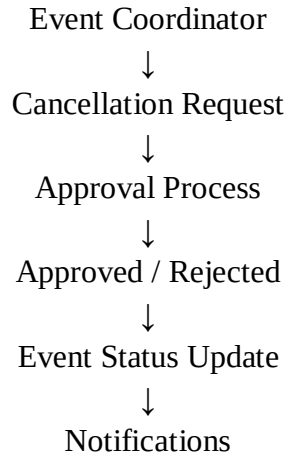
	Reports & Dashboards	
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Main Data Flow:

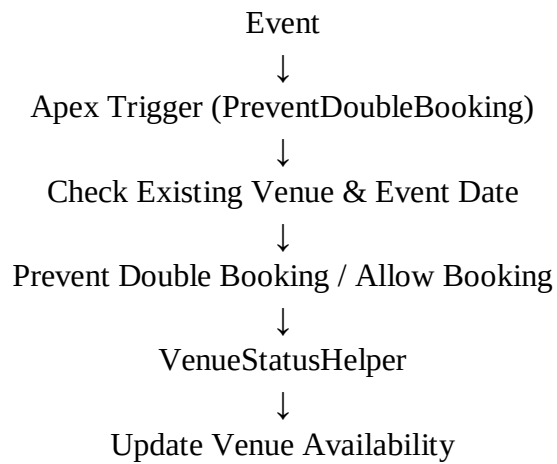


Supporting Flows:

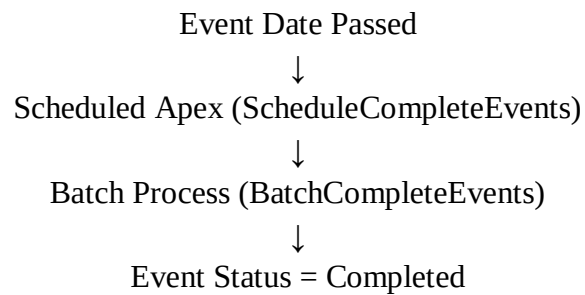
Cancellation Flow:



Venue Booking Validation Flow:



Event Completion Flow:



Reminder Flow:

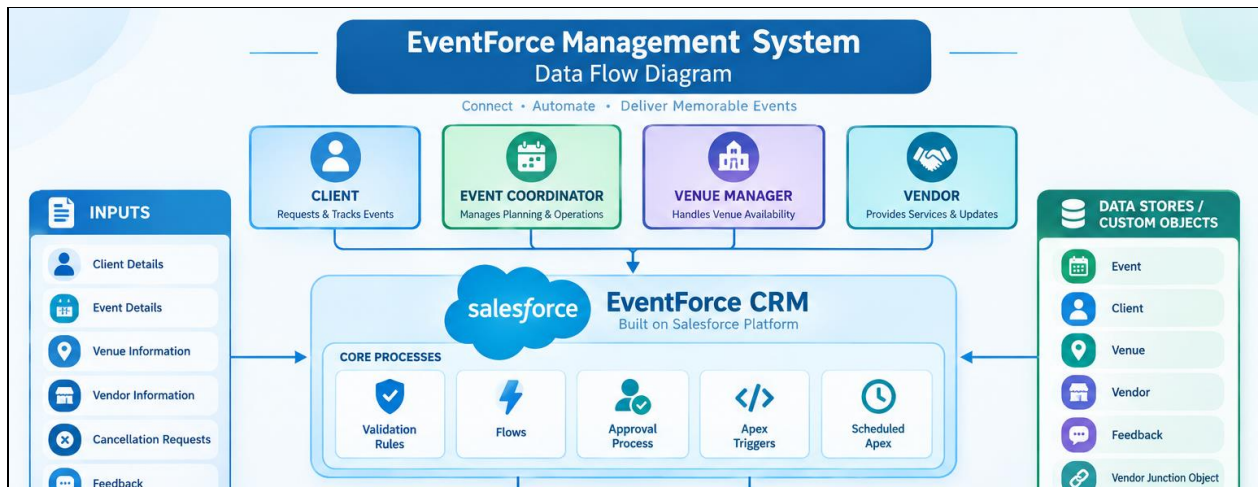
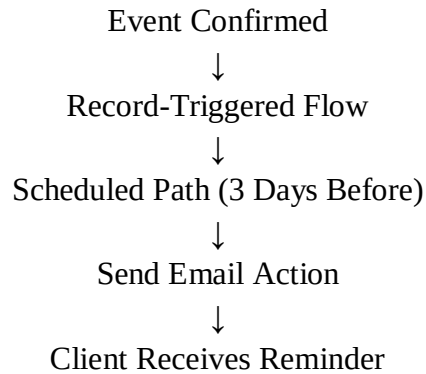


Figure 4: Data Flow Diagram – Main Flow

3.4 Technology Stack

The EventForce Management System is built on Salesforce platform using a combination of declarative and programmatic technologies.

Layer	Technology	Purpose
Platform	Salesforce CRM / Developer Edition	Core platform for application development
UI	Salesforce Lightning App	User interface and navigation
Database	Salesforce Custom Objects	Data storage and management
Business Logic	Salesforce Flows	Declarative automation
	Formula Fields	Calculated values

Layer	Technology	Purpose
	Validation Rules	Data quality enforcement
Approval	Salesforce Approval Process	Cancellation workflow
Automation	Record-Triggered Flows	Event-driven automation
	Apex Triggers	Advanced business logic
Processing	Batch Apex	Bulk data processing
	Scheduled Apex	Time-based execution
Security	Profiles	Object and field permissions
	Roles	Record-level access hierarchy
	Permission Sets	Additional permissions
	Sharing Rules	Record sharing
	OWD	Organization-wide defaults
Reporting	Reports	Data analysis
	Dashboards	Visual monitoring
Data Migration	Data Import Wizard	Sample data loading
Development	Developer Console	Apex development
	Setup	Configuration

Table 6: Technology Stack

Salesforce Features Used:

Feature	Count	Purpose
Custom Objects	6	Data storage
Custom Fields	30+	Data capture
Validation Rules	1	Data quality
Formula Fields	1	Budget calculation
Lookup Filters	1	Relationship filtering
Record-Triggered Flows	1	Reminder automation

Feature	Count	Purpose
Approval Processes	1	Cancellation workflow
Email Templates	4	Notifications
Apex Classes	3	Business logic
Apex Triggers	2	Event-driven logic
Batch Apex	1	Bulk processing
Scheduled Apex	1	Time-based execution
Profiles	5	User permissions
Roles	5	Record hierarchy
Permission Sets	1	Additional access
Sharing Rules	1	Record sharing
Reports	1	Data analysis
Dashboards	1	Visual monitoring
Lightning Apps	1	User interface

4. PROJECT DESIGN PHASE

4.1 Problem-Solution Fit

The EventForce Management System provides a strong fit between identified event-management problems and Salesforce capabilities.

Problem	Solution – EventForce Salesforce System	Fit / Benefit
Client, event, venue, and vendor information is difficult to manage centrally	Custom Salesforce objects for Event, Client, Venue, Vendor, Feedback, and EventVendor	Centralized and structured event data
Event booking requires multiple manual activities	Salesforce Flows and automated processes	Reduces manual effort and improves process efficiency

Problem	Solution – EventForce Salesforce System	Fit / Benefit
Risk of venue double booking	Apex Trigger and VenueStatusHelper	Validates venue availability and prevents double bookings
Vendor coordination is difficult to track	EventVendor junction object and Vendor records	Supports event–vendor assignment and coordination
Clients may not receive timely event reminders	3-Day Reminder Flow	Automates reminder communication before confirmed events
Cancellation requests require controlled processing	Event_Cancellation_Process Approval Process	Provides structured approval and status updates
Client feedback needs to be connected with events	Feedback object with Event and Client relationships	Enables organized feedback management
Operational information is difficult to monitor	Reports and EventForce Operations Dashboard	Provides better visibility for monitoring and decisions
Different users require different levels of access	Profiles, Roles, Permission Sets, OWD, and Sharing Rules	Provides controlled and secure data access
Past events need status updates	Batch Apex and Scheduled Apex	Automatically marks events as Completed

Table 7: Problem-Solution Fit

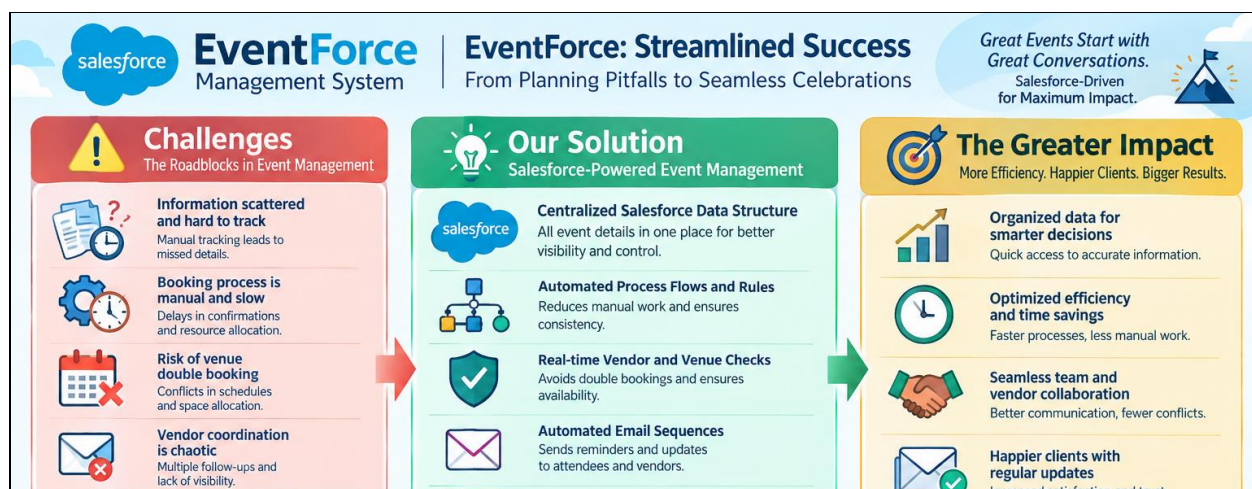


Figure 5: From Challenges to Celebrations – Problem-Solution-Impact

Detailed Fit Analysis:

Centralized

The six custom objects (Event, Client, Vendor, Venue, Feedback, EventVendor) with defined relationships create a comprehensive data model that captures all aspects of event management. Lookup and Master-Detail relationships ensure data integrity and enable cross-object reporting.

Automation:

Record-triggered Flows automate the three-day reminder process, eliminating manual follow-up. The Approval Process automates cancellation routing and notifications. Batch Apex and Scheduled Apex automate event completion.

Conflict

Prevention:

The PreventDoubleBooking Apex trigger checks for existing events at the same venue on the same date before allowing a new booking. This provides a critical layer of protection against scheduling conflicts.

Communication:

Email templates and email alerts ensure timely communication with clients, event owners, and approvers. The reminder Flow sends personalized emails with event details.

Security:

The security model with Profiles, Roles, Permission Sets, OWD, and Sharing Rules ensures that each user sees only the data relevant to their responsibilities.

Visibility:

The Upcoming Events by Month report and EventForce Operations Dashboard provide management with real-time visibility into event schedules and operational metrics.

4.2 Proposed Solution

The proposed solution is a comprehensive Salesforce CRM implementation that addresses all identified business requirements.

Parameter	Description
Problem	Manual and fragmented event management
Solution	Salesforce CRM
Innovation	Automated event reminders, venue validation, and cancellation approvals

Parameter	Description
Impact	Faster event coordination and improved operational visibility
Business Model	Centralized event planning and management
Scalability	Supports growing events, clients, vendors, venues, and users

Table 8: Proposed Solution Parameters

Solution Components:

1. **Data Layer:** Six custom objects with fields and relationships
2. **Automation Layer:** Flows, Approval Processes, Email Alerts, Apex
3. **Security Layer:** Profiles, Roles, Permission Sets, OWD, Sharing Rules
4. **Presentation Layer:** Lightning App, Reports, Dashboards
5. **Integration Layer:** Data Import Wizard for data loading

Key Features:

Feature	Implementation	Benefit
Automated Budget	Formula field based on Event Type	Eliminates manual budget calculation
Email Validation	Validation rule with REGEX	Ensures valid client contact information
Feedback Filter	Lookup filter on Event field	Prevents incorrect event-client associations
Venue Reservation	Apex trigger and helper class	Automatic venue status updates
Double Booking Prevention	Apex trigger	Prevents scheduling conflicts
Cancellation Approval	Approval Process	Controlled cancellation workflow
Client Reminder	Record-triggered Flow	Automated client communication
Event Completion	Batch and Scheduled Apex	Automatic status updates
Role-based Access	Profiles, Roles, Permission Sets	Secure data access
Operational Visibility	Reports and Dashboards	Management insights

4.3 Solution Architecture

The EventForce Management System follows a multi-layer architecture that separates concerns and promotes maintainability.

Architecture Layers:

1. Presentation Layer (UI):

- Salesforce Lightning App
- Role-based dashboards, record pages, lists, and forms
- Navigation menu with tabs for all objects

2. Business Logic & Automation Layer:

- Flows (Automation)
- Validation Rules (Data Quality)
- Approval Process (Cancellations)
- Apex Triggers (Business Logic)
- Batch & Scheduled Apex (Scheduled Jobs)

3. Data Layer (Salesforce Objects):

- Event
- Client
- Venue
- Vendor
- EventVendor (Junction Object)
- Feedback

4. Security & Access Control Layer:

- Profiles
- Roles
- Permission Sets
- Sharing Rules
- OWD

External Communication:

- Email Notifications

- Event Confirmations
- 3-Day Reminders
- Cancellation Updates

Reporting & Analytics:

- Reports & Dashboards
- Event Metrics
- Venue Utilization
- Vendor Performance
- Feedback Insights

Data Management:

- Data Import Wizard
- Import clients, venues, vendors, and event data



Figure 6: EventForce Management System – Solution Architecture

Architecture Principles:

1. **Separation of Concerns:** Each layer has a specific responsibility
2. **Declarative First:** Prefer Flows and validation rules over code where possible
3. **Security by Design:** Access controls at every layer
4. **Scalability:** Design for growth in data volume and users
5. **Maintainability:** Modular components that can be updated independently
6. **Auditability:** Track changes through field history and approval history

4.4 Data Model and Relationships

The data model defines the structure of data storage and the relationships between objects.

Object Details:

Object	Key Fields	Relationship
Event__c	Event Name, Event Date, Event Type, Event Status, Event Budget (formula)	Lookup to Client and Venue
Client__c	Client Name, Email, Phone, Address, Country, City	●
Vendor__c	Vendor Name, Email, Phone, Service Type, Status	●
Venue__c	Venue Name, Address, Location (URL), Capacity, Availability Status	●
Feedback__c	Feedback ID (Auto Number F-{0000}), Rating, Comments	Lookup to Event and Client
EventVendor__c	Junction object	Master-Detail to Event and Vendor

Table 9: Data Model and Relationships

Entity Relationship Diagram:

Object	Fields	Relationships
Client	Client Name, Email, Phone, Address, Country, City	Parent of Event (Lookup) and Feedback (Lookup)
Venue	Venue Name, Address, Capacity, Status	Parent of Event (Lookup)
Event	Event Name, Event Date, Event Type, Event Status, Event Budget (formula)	Client (Lookup), Venue (Lookup)
EventVendor (Junction Object)	—	Event (Master-Detail), Vendor (Master-Detail)
Vendor	Vendor Name, Email, Phone, Service Type, Status	Linked to Event through EventVendor

Object	Fields	Relationships
Feedback	Feedback ID, Rating, Comments	Event (Lookup), Client (Lookup)

Relationship Details:

Parent Object	Child Object	Relationship Type	Field Name
Client	Event	Lookup	Client__c
Venue	Event	Lookup	Venue__c
Event	Feedback	Lookup	Event__c
Client	Feedback	Lookup	Client__c
Event	EventVendor	Master-Detail	Event__c
Vendor	EventVendor	Master-Detail	Vendor__c

Table 10: Relationships Summary

Data Model Design Decisions:

1. **Junction Object:** Salesforce does not support direct many-to-many relationships, so EventVendor serves as a junction object between Event and Vendor
2. **Master-Detail vs Lookup:** Master-Detail used in EventVendor for strong parent-child relationship; Lookup used elsewhere for flexibility
3. **Formula Field:** Event Budget calculated automatically based on Event Type to reduce manual entry
4. **Auto Number:** Feedback ID uses auto-number format F-{0000} for unique identification
5. **Picklist Values:** Predefined values for Event Type, Event Status, Rating, and other fields to ensure consistency

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project followed Agile methodology with Epics – User Stories – Story Points, executed in four sprints. Velocity was used for estimation.

Methodology:

- **Agile Framework:** Iterative development with sprints
- **Epic Breakdown:** Large features broken into user stories
- **Story Points:** Relative estimation of effort
- **Sprint Duration:** 6 days per sprint
- **Velocity Tracking:** Measure of team productivity

Stakeholder Mapping:

Stakeholder	Responsibility
Clients	Request event services, provide feedback, and view event updates
Event Coordinators	Manage event scheduling, client communication, and approval workflows
Vendors	Handle assigned event services like catering, decor, and logistics
Venue Managers	Manage venue availability and reservations
Event Administrators	Oversee system operations, security, and data monitoring
Management	Monitor operations through reports and dashboards

Table 11: Stakeholder Mapping

Project Phases:

Phase	Duration	Key Activities
Phase 1: Requirement Analysis & Planning	Sprint 0	Scope definition, stakeholder mapping, execution roadmap
Phase 2: Backend Development & Configuration	Sprints 1-2	Objects, fields, relationships, validation, flows, approval, Apex
Phase 3: UI/UX Development & Customization	Sprint 3	Lightning App, Reports, Dashboards
Phase 4: Data Migration, Testing & Security	Sprint 4	Data loading, profiles, roles, permission sets, sharing, testing
Phase 5: Deployment, Documentation & Maintenance	Post-Sprint	Documentation, demo video, deployment preparation

5.2 Product Backlog

The product backlog contains all user stories prioritized by business value.

Sprint	Epic ID	User Story	SP	Priority	Member
1	Event Management	US-01: As an Event Coordinator, I want to create events with type, date, and status so that event information is standardized.	3	High	Member 1
1	Client & Venue Management	US-02: As an Event Coordinator, I want to associate a client and venue with an event so that coordination is centralized.	5	High	Member 2
2	Vendor Management	US-03: As a Vendor Manager, I want to assign multiple vendors to an event so that services can be coordinated.	5	High	Member 3
2	Notifications & Automation	US-04: As a Client, I want timely event reminders so that I know when my event is approaching.	3	High	Member 4
3	Approval Management	US-05: As an Event Coordinator, I want cancellation requests to follow an approval process so that cancellations are controlled.	5	High	Member 1
3	Venue Management & Apex	US-06: As a Venue Manager, I want double bookings prevented so that venue schedules remain accurate.	8	High	Member 2
3	Security & Access Control	US-07: As an Event Administrator, I want role-based access so that users see only appropriate information.	5	High	Member 3
4	Reports & Dashboards	US-08: As Management, I want reports and dashboards so that event operations can be monitored.	5	Medium	Member 4

Table 12: Product Backlog

User Story Details:

US-01: Event Creation

- **As an** Event Coordinator
- **I want to** create events with type, date, and status
- **So that** event information is standardized
- **Acceptance Criteria:**
 - Event record can be created with required fields
 - Event Type picklist includes Wedding, Corporate, Birthday, Anniversary, Festival, Concert, Other
 - Event Date field accepts valid dates
 - Event Status picklist includes Planned, Confirmed, Completed, Pending Cancellation, Canceled, Rejected
 - Event Budget calculated automatically based on Event Type

US-02: Client and Venue Association

- **As an** Event Coordinator
- **I want to** associate a client and venue with an event
- **So that** coordination is centralized
- **Acceptance Criteria:**
 - Event record has lookup fields for Client and Venue
 - Client lookup shows all client records
 - Venue lookup shows all venue records
 - Relationships are saved correctly

US-03: Vendor Assignment

- **As a** Vendor Manager
- **I want to** assign multiple vendors to an event
- **So that** services can be coordinated
- **Acceptance Criteria:**
 - EventVendor junction object allows multiple vendor associations
 - Master-Detail relationships ensure data integrity
 - Vendor details accessible from event record

US-04: Event Reminders

- **As a** Client
- **I want** timely event reminders

- **So that** I know when my event is approaching
- **Acceptance Criteria:**
 - Flow sends email 3 days before Event Date
 - Email contains event name, date, venue, and type
 - Email sent to Client email address
 - Event Owner CC'd on email

US-05: Cancellation Approval

- **As an** Event Coordinator
- **I want** cancellation requests to follow an approval process
- **So that** cancellations are controlled
- **Acceptance Criteria:**
 - Approval process triggered when status = Pending Cancellation
 - Request routed to manager of record submitter
 - Approver can approve or reject
 - Status updated based on decision
 - Email notifications sent

US-06: Double Booking Prevention

- **As a** Venue Manager
- **I want** double bookings prevented
- **So that** venue schedules remain accurate
- **Acceptance Criteria:**
 - Apex trigger checks existing events
 - Same venue and same date blocked
 - Error message displayed to user
 - Canceled and Rejected events excluded

US-07: Role-Based Access

- **As an** Event Administrator
- **I want** role-based access
- **So that** users see only appropriate information
- **Acceptance Criteria:**
 - Profiles configured for each role
 - Object permissions set appropriately
 - OWD set to Private for Event
 - Sharing rules extend access where needed

US-08: Reports and Dashboards

- **As** Management
- **I want** reports and dashboards
- **So that** event operations can be monitored
- **Acceptance Criteria:**
 - Report shows upcoming events by month
 - Dashboard displays report data
 - Budget totals calculated
 - Canceled events excluded

5.3 Sprint Schedule

Sprint	Total SP	Duration	Start Date	End Date (Planned)	SP Completed	Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022	20	05 Nov 2022
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022	20	12 Nov 2022
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022	20	19 Nov 2022

Table 13: Sprint Schedule

Sprint Goals:

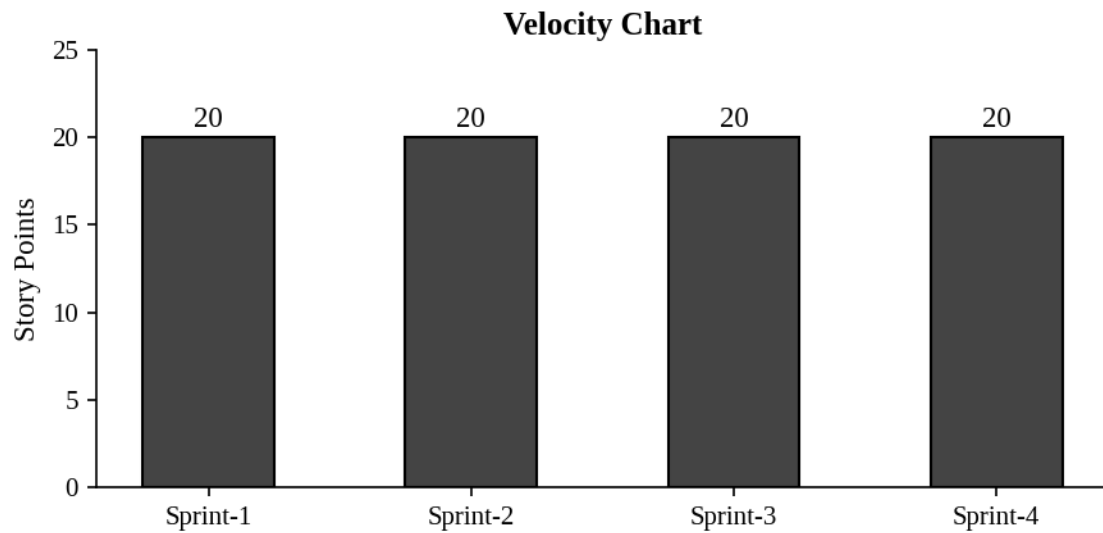
Sprint	Goal	Deliverables
Sprint-1	Foundation setup	Developer account, custom objects, tabs, fields, relationships
Sprint-2	Automation implementation	Validation rules, formula fields, flows, approval process
Sprint-3	Advanced logic and UI	Apex classes, triggers, batch Apex, Lightning App
Sprint-4	Security and analytics	Profiles, roles, permission sets, sharing, reports, dashboards

5.4 Velocity Calculation

Average Velocity: $80 \text{ story points} \div 4 \text{ sprints} = 20 \text{ story points per sprint}$

Daily Velocity: $20 \text{ story points} \div 6 \text{ days} = \text{approximately } 3.33 \text{ points per day}$

Velocity Chart:



Estimation Accuracy:

Sprint	Planned SP	Completed SP	Accuracy
Sprint-1	20	20	100%
Sprint-2	20	20	100%
Sprint-3	20	20	100%
Sprint-4	20	20	100%
Total	80	80	100%

6. PROJECT DEVELOPMENT PHASE

6.1 Phase 1: Requirement Analysis & Planning

Scope: Custom objects (Event, Client, Vendor, Venue, Feedback, EventVendor); relationships (Lookup, Master-Detail, Many-to-Many); Flow automation; validation rules and formula fields; approval process with email alerts; profiles, roles, and permission sets; Apex classes and triggers; and reports and dashboards.

Functional Scope:

Area	Components
Custom Objects	Event, Client, Vendor, Venue, Feedback, Event Vendor (Junction Object)
Relationships	Lookup, Master-Detail, Many-to-Many (Event – Vendor)
Automation	Flows for reminders, feedback notifications, approval updates
Data Quality	Validation Rules, Formula Fields for budget calculation
Approval	Approval Processes for event cancellation with email alerts
Security	Profiles, Roles, Permission Sets for user-level access
Analytics	Reports and Dashboards for event summary and insights

Stakeholder Mapping:

Stakeholder	Responsibility
Clients	Request event services, provide feedback, and view event updates
Event Coordinators	Manage event scheduling, client communication, and approval workflows
Vendors	Handle assigned event services like catering, decor, and logistics
Venue Managers	Manage venue availability and reservations
Event Administrators	Oversee system operations, security, and data monitoring

Execution Roadmap:

Step	Activity	Deliverable
1	Object & Field Setup	Creation of Event, Client, Vendor, Venue, Feedback, and Event Vendor objects
2	Automation Flows	Record-triggered flows for event reminders and approvals
3	Validation Rules & Formula Fields	Data accuracy and budget calculations
4	Profiles, Roles, and Permission Sets	User access and record visibility
5	Approval Process	Event cancellation request workflow with notifications
6	Apex Classes & Triggers	Venue availability and double booking prevention
7	Reports & Dashboards	Analytics for upcoming events and performance insights

6.2 Phase 2: Backend Development & Configuration

The Backend & Configuration phase built the core functional engine of the EventForce Management System. Using a combination of Salesforce Flows, Apex logic, and Security configurations, this phase focused on developing the system's automation backbone to manage events, clients, vendors, venues, and feedback efficiently.

Declarative tools such as Record-Triggered Flows, Validation Rules, Formula Fields, and Approval Processes were used extensively to automate booking confirmations, reminders, and cancellation approvals. Meanwhile, Apex Classes and Triggers handled advanced business logic like venue availability updates and preventing double bookings, ensuring operational accuracy and scalability.

The phase also included implementing Profiles, Roles, Permission Sets, and Sharing Rules to define user-level access, maintaining data integrity and security across all modules.

6.2.1 Milestone 1: Salesforce Developer Account

What is Salesforce?

Salesforce is a customer success platform designed to help organizations sell, service, market, analyze, and connect with customers. Salesforce provides everything needed to run a business from anywhere. Using standard products and features, organizations can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store data securely in the cloud.

Activity 1: Creating Developer Account

1. Go to <https://developer.salesforce.com/signup>
2. On the sign-up form, enter the following details:
 - First name & Last name
 - Email
 - Role: Developer
 - Company: College Name
 - Country: India
 - Postal Code: PIN code
 - Username: Should be a combination of name and company (format: username@organization.com)
3. Click on "Sign me up" after filling these details

Build for the Agentic Enterprise. Sign up for your Developer Edition.

The entire Salesforce platform – expressed as APIs, MCP tools, and CLI commands. Build apps, ship agents, and let coding agents do the heavy lifting with Salesforce Headless 360.

- ✓ Build AI agents with Agentforce, grounded in your data with Data 360
- ✓ Connect Claude Code, Cursor, or Codex to your org with hosted MCP
- ✓ Build in the browser with the Agentforce Vibes IDE - no setup required
- ✓ Drag-and-drop when you want it, Apex when you need it
- ✓ 60+ MCP tools and 30+ coding skills, ready out of the box

Sign up for your Developer Edition
A free Salesforce Platform environment with Agentforce and Data 360

First name* Mythili ✓ Last name* S ✓

Job title* Student ✓ Company* Vivekanandha college of te... ✓

Country/Territory* India ✓ Work email* MaithiliMalar006@gmail.com ✓

☒ I agree to the Main Services Agreement, Developer Products, Salesforce Program, Agreement, and APAC Terms of Service.

Your free trial may be terminated at any time if you are not a student, developer, or teacher. Salesforce, and its affiliates, reserve the right to limit, suspend, or terminate any free trial at any time in their sole discretion, with or without notice.

We value your privacy. To learn more, view our [Privacy Statement](#).

Sign me up

Figure 7: Salesforce Developer Edition Signup Page

Activity 2: Account Activation

1. Go to the inbox of the email used while signing up
2. Click on "Verify Account" to activate the account (email may take 5-10 minutes)
3. Give a password and answer a security question
4. Click on "Change Password"
5. Redirect to Salesforce Setup page

Figure 8: Salesforce Account Verification Email

6.2.2 Milestones 2-4: Objects, Tabs, Fields, Relationships

What is an Object?

Salesforce objects are database tables that permit storage of data specific to an organization. They consist of fields (columns) and records (rows).

Data Architecture:

Created Custom Objects:

Object	API Name	Fields
Event	Event__c	Event Name, Event Date, Event Type, Event Status, Event Budget
Client	Client__c	Client Name, Email, Phone, Address, Country, City
Vendor	Vendor__c	Vendor Name, Email, Phone, Service Type, Status
Venue	Venue__c	Venue Name, Address, Location (URL), Capacity, Availability Status
Feedback	Feedback__c	Feedback ID (Auto Number), Rating, Comments
Event Vendor	EventVendor__c	Junction Object (Links Event – Vendor)

Activity 1: Create Event Object

1. From Setup, click on Object Manager
2. Click Create, select Custom Object
3. Fill in the label as "Event"
4. Fill in the plural label as "Events"
5. Record name: "Event Name"
6. Select the data type as "Text"
7. In the Optional Features section, select Allow Reports, Allow Activities, and Track Field History
8. In the Deployment Status section, ensure Deployed is selected
9. In the Search Status section, select Allow Search
10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout
11. Click Save

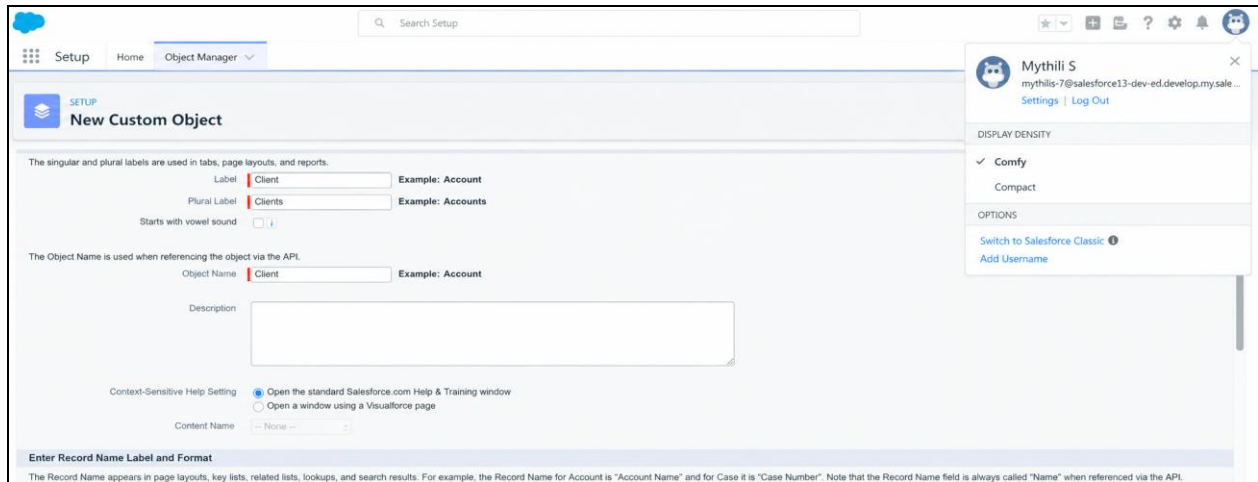


Figure 9: Object Manager showing the deployed custom objects

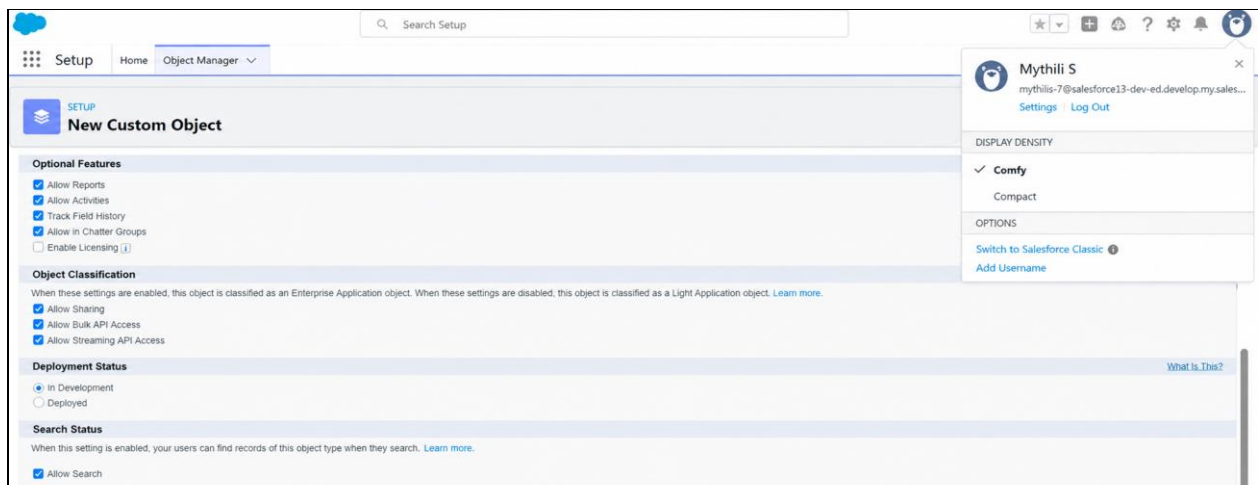


Figure 10: New Custom Object Definition Page

Activity 2: Create Client Object

1. Click on the gear icon and select Setup
2. Click on the Object Manager tab
3. Click Create dropdown and select Custom Object
4. On the Custom Object Definition page:
 - Label: Client
 - Plural Label: Clients
 - Record Name: Client Name
 - Data Type: Text
 - Check Allow Reports
 - Check Allow Search

- Select Add Notes and Attachments related list to default page layout
- 5. Click Save

Activity 3: Create Vendor Object

1. Navigate to Object Manager – Create – Custom Object
2. Configure:
 - Label: Vendor
 - Plural Label: Vendors
 - Record Name: Vendor Name
 - Data Type: Text
 - Check Allow Reports
 - Check Allow Search
3. Click Save

Activity 4: Create Venue Object

1. Navigate to Object Manager – Create – Custom Object
2. Configure:
 - Label: Venue
 - Plural Label: Venues
 - Record Name: Venue Name
 - Data Type: Text
 - Check Allow Reports
 - Check Allow Search
3. Click Save

Activity 5: Create Feedback Object

1. Navigate to Object Manager – Create – Custom Object
2. Configure:
 - Label: Feedback
 - Plural Label: Feedbacks
 - Record Name: Feedback
 - Data Type: Auto Number
 - Display Format: F-{0000}
 - Starting Number: 0
 - Check Allow Reports
 - Check Allow Search
3. Click Save

Milestone 3: Tabs

What are Tabs?

Tabs in Salesforce help users view information at a glance. They display data of objects and other web content in the application.

Types of Tabs:

- **Standard Object Tabs:** Display data related to standard objects
- **Custom Object Tabs:** Display data related to custom objects (look and function like standard tabs)

Activity 1: Creating a Custom Tab (Event)

1. Click on the Home tab, enter "Tabs" in Quick Find, and select Tabs
2. Under Custom Object Tabs, click New
3. For Object, select Event
4. For Tab Style, select any icon
5. Leave all defaults as is
6. Click Next, Next, and Save

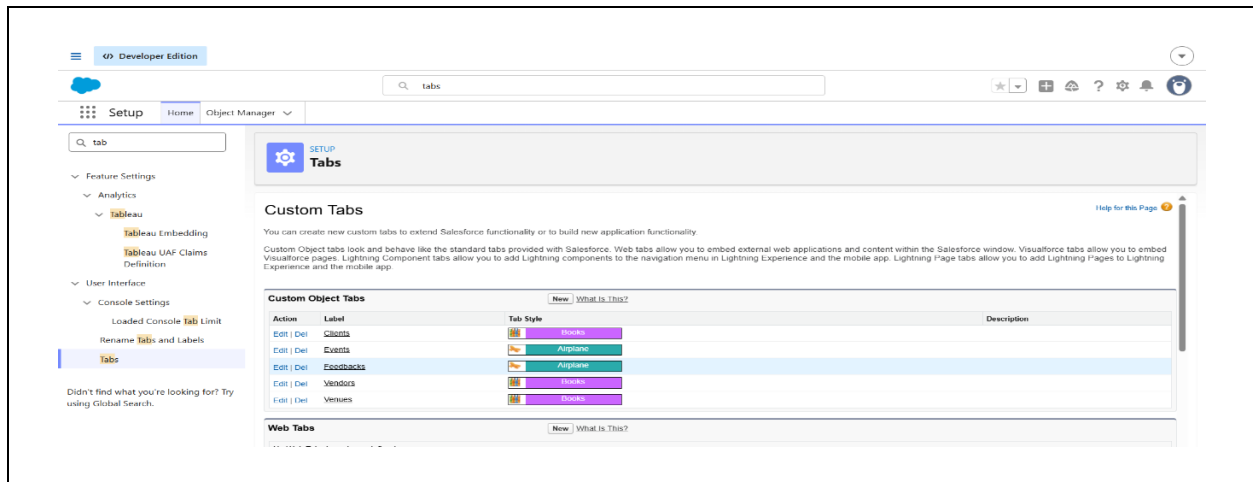


Figure 11: Custom Tabs Creation Page

Activity 2-5: Creating Custom Tabs for Client, Vendor, Venue, and Feedback

Repeat the same steps as above, selecting the respective object for each tab.

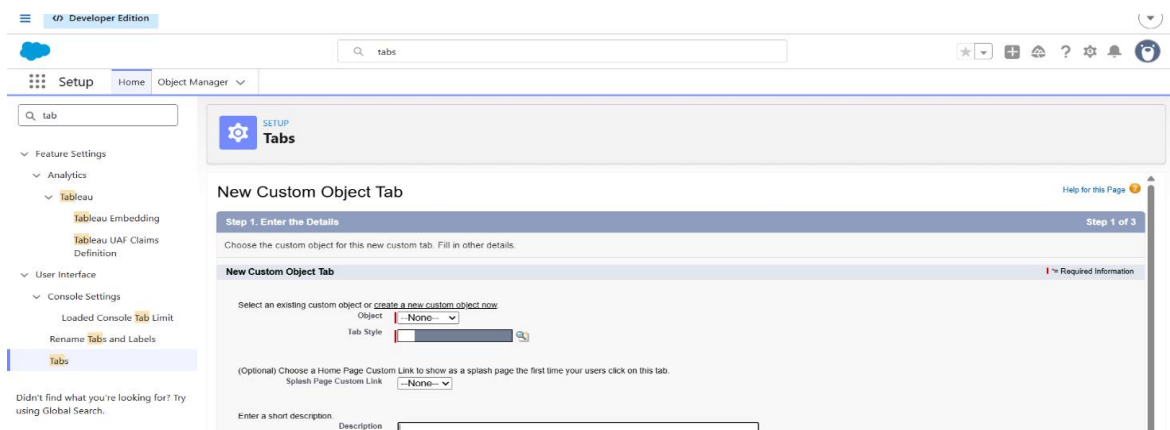
Milestone 4: Fields and Relationships

What are Fields?

Fields in Salesforce represent columns in relational databases. They store data values required for a particular object in a record.

Types of Fields:

- **Standard Fields:** Created By, Last Modified By, Owner, and the field created at object creation. Cannot be deleted or edited.
- **Custom Fields:** Added by administrators/developers to meet business requirements.



Activity 1: Creation of Fields for Event Object

1. Click the gear icon and select Setup
2. Click the Object Manager tab
3. Select Event
4. Select Fields & Relationships from the left navigation, and click New
5. Choose the data type as Picklist, click Next
6. For Field Label, enter "Event Type"
7. Select "Enter values, with each value separated by a new line" and enter:
 - Wedding
 - Corporate
 - Birthday
 - Anniversary
 - Festival
 - Concert
 - Other
8. Click Next, Next, and Save

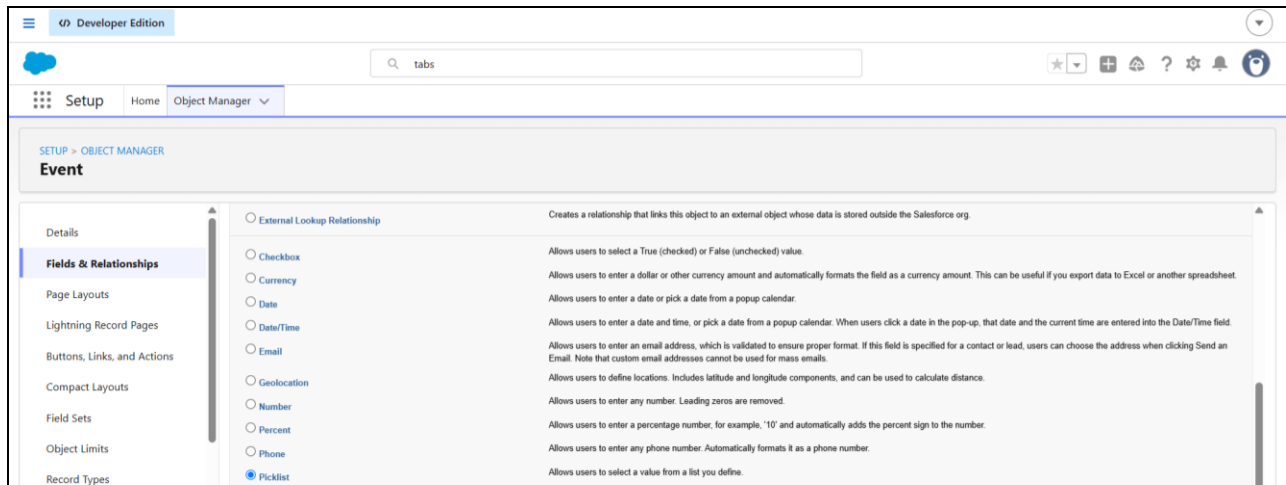


Figure 12: Event Type Picklist Field Creation

Creating Additional Event Fields:

1. **Event Date:** Select Date as Data Type, Field Label "Event Date"
2. **Event Status:** Select Picklist as Data Type, Field Label "Event Status", Enter Values:
 - Planned
 - Confirmed
 - Completed
 - Pending Cancellation
 - Canceled
 - Rejected

Creating Formula Field for Event Budget:

1. Select Data type as "Formula" and click Next
2. Enter field label "Event Budget"
3. Select formula return type "Currency", click Next
4. Create and insert formula:

```
CASE(
  TEXT(Event_Type__c),
  "Wedding", 50000,
  "Corporate", 30000,
  "Birthday", 10000,
  "Anniversary", 20000,
  "Festival", 60000,
  "Concert", 40000,
```

"Other", 15000,
0
)

5. Check the syntax
6. Click Next, Next, and Save

The screenshot shows the Salesforce Setup interface in the 'Developer Edition'. The breadcrumb trail is 'SETUP > OBJECT MANAGER'. The main heading is 'Event'. On the left sidebar, 'Fields & Relationships' is selected. The main content area is titled 'Step 2. Enter the details'. It shows a 'Field Label' of 'Event Type'. Under 'Values', the option 'Enter values, with each value separated by a new line' is selected. A text area contains a list of event types: Wedding, Corporate, Birthday, Anniversary, Festival, Concert, and Other. At the bottom, there is a checkbox for 'Display values alphabetically, not in the order entered' which is currently unchecked. Navigation buttons 'Previous', 'Next', and 'Cancel' are visible in the top right of the configuration area.

Figure 13: Event Budget Formula Field Definition

Activity 2: Creation of Fields for Client Object

1. **Email:** Select Email as Data Type, Field Label "Email"
2. **Phone:** Select Phone as Data Type, Field Label "Phone"
3. **Address:** Select Text Area as Data Type, Field Label "Address"
4. **Country:** Select Picklist as Data Type, Field Label "Country", Enter Values:
 - India, Philippines, Saudi Arabia, USA, Switzerland, Thailand, Maldives, Dubai, Sri Lanka, Bangkok, Singapore, Myanmar, France, Nepal
5. **City:** Select Picklist as Data Type, Field Label "City", Enter Values:
 - Hyderabad, Mumbai, Delhi, Bangalore, Chennai, Alaminos, Angeles City, Palayan, Quezon City, San Juan, Riyadh, Jeddah, Tayma, Dammam, Medina, New York, Los Angeles, Chicago, Houston, Columbus, Geneva, Basel, Lausanne, Bern, Lucerne

Activity 3: Creation of Fields for Vendor Object

1. **Service Type:** Select Multi-Select Picklist as Data Type, Field Label "Service Type", Enter Values:

- Catering, Decor, Photography, Videography, Lighting, Stage Setup, Makeup Artist, DJ/Music, Transportation, Hosting/Anchor
- 2. **Email:** Select Email as Data Type, Field Label "Email"
- 3. **Phone:** Select Phone as Data Type, Field Label "Phone"
- 4. **Status:** Select Picklist as Data Type, Field Label "Status", Enter Values:
 - Available, Booked, Cancelled

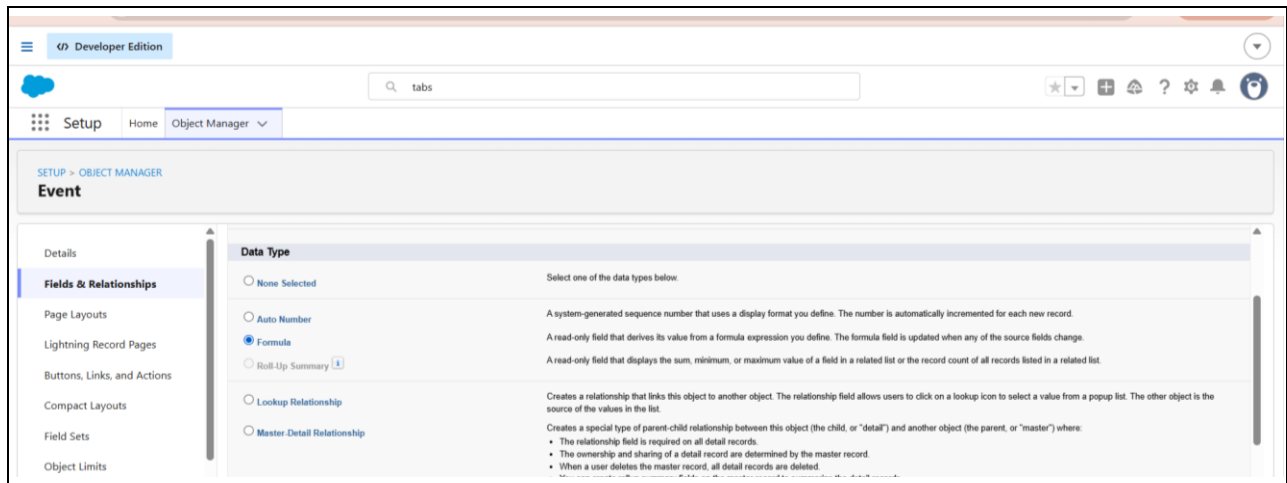


Figure 15: Vendor Object Fields

Activity 4: Creation of Fields for Venue Object

1. **Address:** Select Text Area as Data Type, Field Label "Address"
2. **Location:** Select URL as Data Type, Field Label "Location"
3. **Capacity:** Select Number as Data Type, Field Label "Capacity"
4. **Availability Status:** Select Picklist as Data Type, Field Label "Availability Status", Enter Values:
 - Available, Reserved, Not Available

Activity 5: Creation of Fields for Feedback Object

1. **Rating:** Select Picklist as Data Type, Field Label "Rating", Enter Values:
 - 1, 2, 3, 4, 5
2. **Comments:** Select Long Text as Data Type, Field Label "Comments"

Relationships

Activity 1: Creation of Lookup Relationship for Event Objects

Creating Lookup Relationship (Event – Venue):

1. From Setup, go to Object Manager, Select Event Object
2. On the sidebar, click Fields & Relationships
3. Select Lookup Relationship & click Next
4. Choose the related object as Venue & click Next
5. Give the field label (Venue) & click Next, Next, Next, and Save

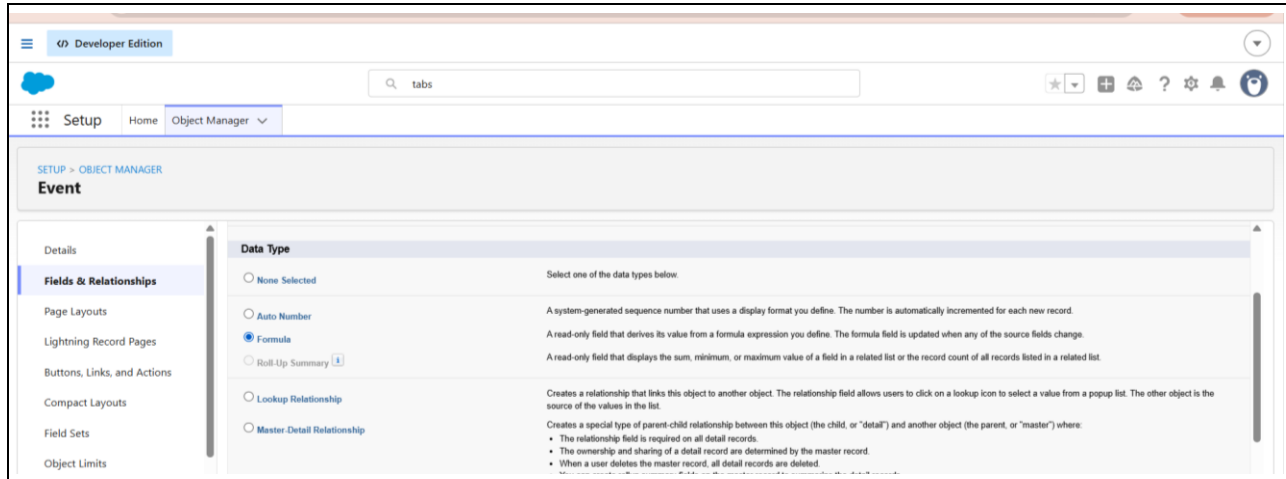


Figure 18: Event to Venue Lookup Relationship

Creating Lookup Relationship (Event – Client):

1. From Setup, go to Object Manager, Select Event Object
2. On the sidebar, click Fields & Relationships
3. Select Lookup Relationship & click Next
4. Choose the related object as Client & click Next
5. Give the field label (Client) & click Next, Next, Next, and Save

Creating Many-to-Many Relationship (Event – Vendor):

Salesforce doesn't allow direct many-to-many, so a junction object (EventVendor__c) is created:

1. Go to Object Manager – Create New Custom Object
 - Label: Event Vendor
 - Object Name: EventVendor__c
2. From Setup, go to Object Manager, Select Event Vendor Object
3. On the sidebar, click Fields & Relationships
4. Select Master-Detail Relationship & click Next
5. Choose the related object as Event & click Next

6. Give the field label (Event) & click Next, Next, Next, and Save & New
7. Select Master-Detail Relationship & click Next
8. Choose the related object as Vendor & click Next
9. Give the field label (Vendor) & click Next, Next, Next, and Save

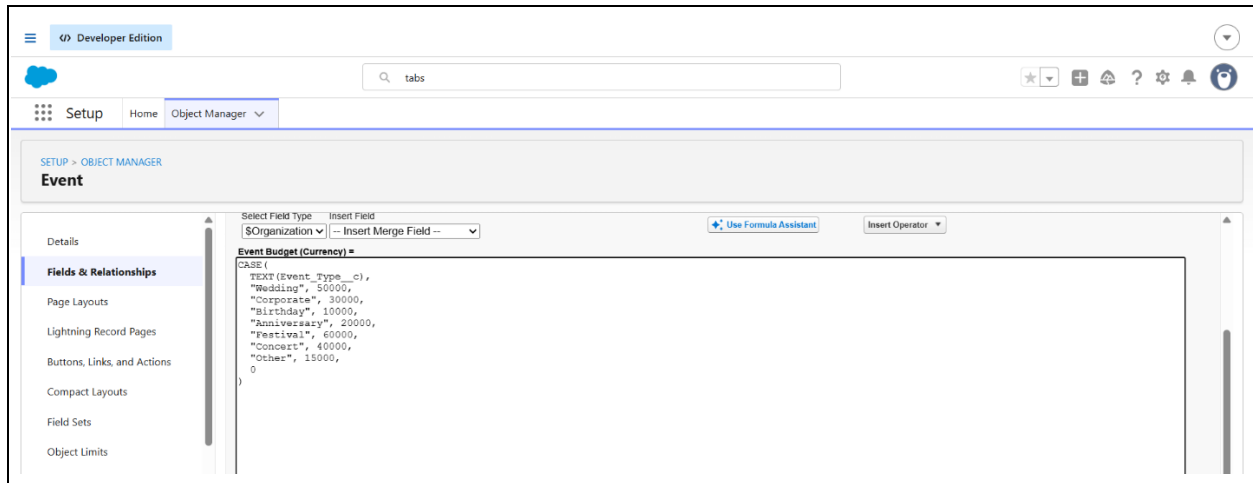


Figure 19: EventVendor Junction Object with Master-Detail Relationships

Activity 2: Creation of Lookup Relationship for Feedback Object

Creating Lookup Relationship (Feedback → Event):

1. From Setup, go to Object Manager, Select Feedback Object
2. On the sidebar, click Fields & Relationships
3. Select Lookup Relationship & click Next
4. Choose the related object as Event & click Next
5. Give the field label (Event) & click Next, Next, Next, and Save

Creating Lookup Relationship (Feedback – Client):

1. From Setup, go to Object Manager, Select Feedback Object
2. On the sidebar, click Fields & Relationships
3. Select Lookup Relationship & click Next
4. Choose the related object as Client & click Next
5. Give the field label (Client) & click Next, Next, Next, and Save

6.2.3 Milestone 5: Validation Rule and Lookup Filter

What are Validation Rules?

Validation rules are applied when a user tries to save a record and are used to check if the data meets specified criteria. If the criteria are not met, the validation rule triggers an error message and prevents the user from saving the record until the issues are resolved.

Activity 1: Create Validation Rules on Client__c

Scenario: Ensure that the client email follows a valid email format.

1. Go to Setup – click on Object Manager – type "Client" in Quick Find box, click on the Client object
2. Click on Validation Rules – click New
3. Enter the Rule Name as "Email_Valid_Address"
4. Select Active
5. Insert the Error Condition Formula:

`NOT(REGEX(Email__c, "^[a-zA-Z0-9._]+@[a-zA-Z0-9.]+\.[a-zA-Z]{2,}$"))`

6. Enter the Error Message as "Please Enter Valid Email Address"
7. Select the Error Location as Field and select the field as "Email"
8. Click Save

Activity 2: Lookup Filter

Scenario: In an Event Management app, each Event belongs to a Client. When creating Feedback, after selecting a Client, the Event lookup should only show Events of that Client.

1. Click the gear – Setup
2. In Setup Quick Find type Object Manager – open Object Manager
3. Find and open the Feedback object (Feedback__c)
4. In the left menu click Fields & Relationships
5. Find the Event lookup field (Event__c) and click it
6. Click On Edit
7. Scroll to the Lookup Filter section and click Show filter Settings
8. Build the filter criteria:
 - Field (left side): choose the Event object's client field – Event: Client__c
 - Operator: equals
 - Value (right side): choose the Field radio button and then select the Client field on the Feedback record – Feedback: Client__c
9. Make sure to Check "Enable this Filter" box
10. Click on Save

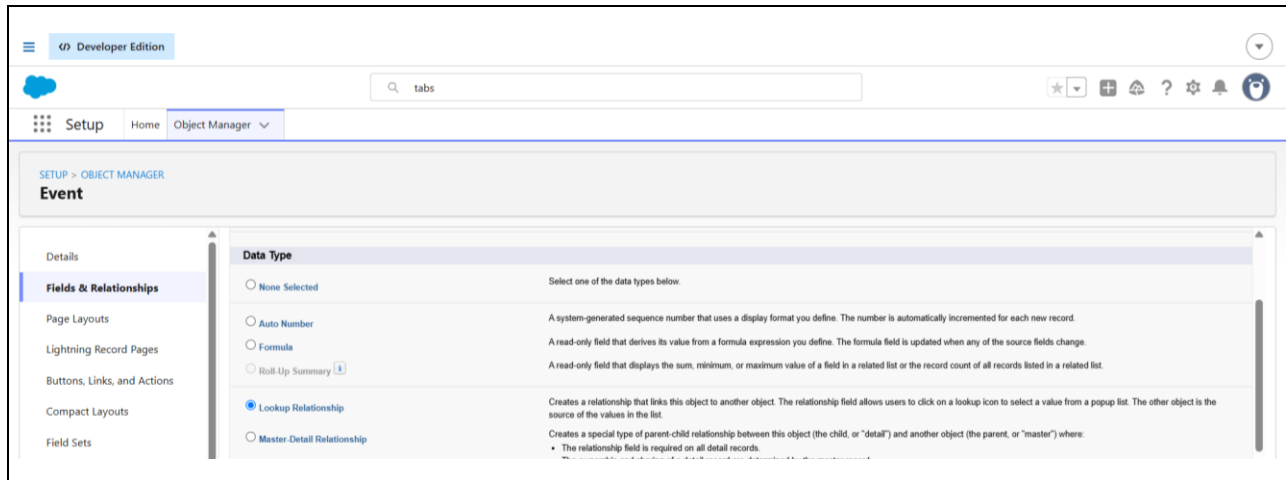


Figure 21: Event Lookup Filter

6.2.4 Milestone 6: Approval Process (Event Cancellation)

Activity 1: Create an Approval Process for Event Cancellation Approval

Note: Before implementing the Approval Process, complete Profiles, Roles, and Users milestones.

Step 1: Create Classic Email Templates

Email Template 1: Event Cancellation Request Notification

1. Go to Setup (gear icon on top-right)
2. In the Quick Find box, type: Email Templates
3. Click on Classic Email Templates
4. Click "New Template"
5. Choose the type: Text – Plain text (no formatting), Click Next
6. Email Template Information:
 - Folder: Unfiled Public Email Templates
 - Email Template Name: Event Cancellation Request Notification
 - Subject: Approval Request: Cancel Event {!Event__c.Name}
 - Email Body:

Hello!

An approval request has been submitted to cancel the event:

- Event Name: {!Event__c.Name}
- Event Date: {!Event__c.Event_Date__c}

Thank you,

EventForce Management System

Email Template 2: Email to Client After Cancellation

- Template Name: Email to Client After Cancellation
- Subject: Event Cancellation Notice – {!Event__c.Name}
- Body:

Hello!

We would like to inform you that the event has been canceled.

Event Name: {!Event__c.Name}

Date: {!Event__c.Event_Date__c}

If you have any questions or need assistance, please contact us.

Thank you for your understanding,
Event Management Team

Email Template 3: Approval Notification to Event Owner

- Template Name: Approval Notification to Event Owner
- Subject: Event Cancellation Approved: {!Event__c.Name}
- Body:

Hello {!Event__c.OwnerFullName}

Your cancellation request for the event {!Event__c.Name} has been approved.

Event Date: {!Event__c.Event_Date__c}

The event status has been updated to Canceled.

Thank you,
EventForce Management System

Email Template 4: Rejection Notification to Event Owner

- Template Name: Rejection Notification to Event Owner
- Subject: Event Cancellation Rejected: {!Event__c.Name}
- Body:

Hello {!Event__c.OwnerFullName},

Your cancellation request for the event {!Event__c.Name} has been rejected.

Event Date: {!Event__c.Event_Date__c}

The event status has been reverted to Confirmed.

Thank you,
EventForce Management System

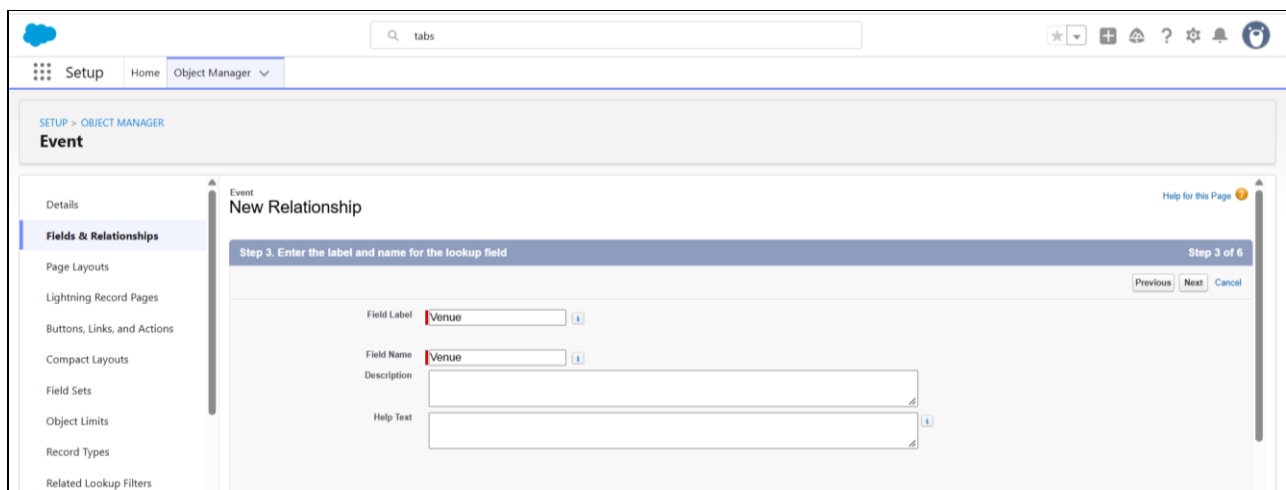


Figure 22: Classic Email Templates for Cancellation Process

Step 2: Create the Approval Process

1. In the Quick Find box, type: Approval Processes
2. Click on "Approval Processes" under Process Automation
3. Click on the object name: Event
4. Click Create New Approval Process – Use Standard Setup Wizard

Setup Home Object Manager

SETUP New Custom Object

Help for this Page

Permissions for this object are disabled for all profiles by default. You can enable object permissions in permission sets or by editing custom profiles. [Tell me more!](#) [Don't show this message again](#)

Custom Object Definition Edit Save Save & New Cancel

Custom Object Information

The singular and plural labels are used in tabs, page layouts, and reports.

Label Example: Account

Plural Label Example: Accounts

Starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name Example: Account

Description

Figure 23: Approval Processes Setup Page

Step 1: Basic Settings

- Name: Event_Cancellation_Process
- Unique Name: Auto-fills
- Click Next

Setup Home Object Manager

Search Setup

Classic Email Templates

Unfiled Public Classic Email Templates

Classic Email Template Availability [Expanded]

Folder: Unfiled Public Classic Email Templates Create New Folder

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Other All

Action	Email Template Name	Template Type	Available For Use	Description	Author	Last Modified Date
Edit Del	Appointment for Unauthenticated User using Appointment Types - For Amazon Chime.	Custom	✓	Email template for confirmation of an appointment when appointments are scheduled using appointment types with Amazon Chime.	sttcadm	9/23/2026
Edit Del	Appointment for Unauthenticated User using Appointment Types - For third party.	Custom	✓	Email template for confirmation of an appointment when appointments are scheduled using appointment types with third party video applications.	sttcadm	9/23/2026
Edit Del	Appointment for Unauthenticated User using Engagement Channels For Amazon Chime.	Custom	✓	Email template for confirmation of an appointment when appointments are scheduled using engagement channels with Amazon Chime.	sttcadm	9/23/2026
Edit Del	Appointment for Unauthenticated User using Engagement	Custom	✓	Email template for confirmation of an appointment when appointments are scheduled using	sttcadm	9/23/2026

Figure 24: Approval Process Basic Settings

Specify Entry Criteria:

- Choose "Criteria are met"
- Set:
 - Field: Event – Event Status
 - Operator: Equals

- Value: Pending Cancellation
- Click Next

Figure 25: Approval Process Entry Criteria

Step 2: Select Approver

- Select: Automatically assign to the manager of the user who submitted the record
- (Ensure users have a manager assigned in their User record)
- Click Next



Figure 26: Select Approver Configuration

Step 3: Email Template

- Approval Assignment Email Template: Event Cancellation Request Notification
- Click Next

Step 4: Select Fields to Display on Approval Page Layout

Choose the fields approvers should see:

- Event Name
- Owner
- Client
- Last Modified By
- Created By

- Event Budget
- Event Date
- Event Status
- Event Type
- Venue

Use the Add button to move fields from left to right. Click Next.

The screenshot shows the 'Approval Processes' setup page in Salesforce. The page is titled 'Approval Processes' and is part of the 'Setup' menu. It is currently on 'Step 3 of 6: Specify Approver Field and Record Editability Properties'. The page includes a search bar at the top, a 'Manager' dropdown, and a 'Help for this Page' link. The main content area is divided into two sections: 'Select Field Used for Automated Approval Routing' and 'Record Editability Properties'. In the first section, the 'Next Automated Approver Determined By' dropdown is set to 'Manager', and the 'Use Approver Field of Event Owner' checkbox is unchecked. In the second section, the 'Administrators ONLY can edit records during the approval process' radio button is selected. Navigation buttons 'Previous', 'Next', and 'Cancel' are located at the bottom right of the page.

Figure 27: Fields to Display on Approval Page Layout

Step 5: Specify Initial Submitters

- Select Event Owner as Initial Submitters
- Click Save

The screenshot shows the 'Approval Processes' setup page in Salesforce, specifically the 'Specify Initial Submitters' step. The page is titled 'Approval Processes' and is part of the 'Setup' menu. It includes a search bar at the top, a 'Home' button, and an 'Object Manager' dropdown. The main content area is divided into two sections: 'Initial Submitters' and 'Available Submitters'. In the 'Initial Submitters' section, the 'Submitter Type' dropdown is set to 'Owner', and the 'Search' field contains 'Owner'. The 'Find' button is visible. In the 'Available Submitters' section, there are two lists: 'Available Submitters' (containing 'None') and 'Allowed Submitters' (containing 'Event Owner'). Between the lists are 'Add' and 'Remove' buttons. Navigation buttons 'Previous', 'Save', and 'Cancel' are located at the top right of the page.

Figure 28: Initial Submitters Configuration

Step 6: Initial Submission Actions

Click Add New – Email Alert

- Description: Alert - Event Cancellation Request
- Unique Name: Auto-filled
- Object: Event__c
- Email Template: Event Cancellation Request Notification
- Recipients: Add Role - Event Coordinator
- From Email Address: Select Current User's Email Address
- Click Save

The screenshot shows the 'Approval Processes' setup page for 'Event_Cancellation_Process'. The page has a header with a gear icon and the word 'SETUP'. Below the header, the page title is 'Approval Processes' and the specific process is 'Event.: Event_Cancellation_Process'. There is a 'Help for this Page' link with a question mark icon. The main content area is titled 'Process Definition Detail' and includes buttons for 'Edit', 'Clone', and 'Delete'. The process details are as follows:

Process Name	Event_Cancellation_Process	Active	☐ i
Unique Name	Event_Cancellation_Process	Next Automated Approver Determined By	Manager of Record Submitter
Description			
Entry Criteria	Event.: Event Status EQUALS Pending Cancellation		
Record Editability	Administrator OR Current Approver	Allow Submitters to Recall Approval Requests	☐
Approval Assignment Email Template	Event_Cancellation Request Notification		
Initial Submitters	Event Owner		
Created By	Sushanth Sush	9/22/2025, 11:12 PM	Modified By Sushanth Sush 9/23/2025, 7:14 AM

Figure 29: Initial Submission Email Alert Configuration

Step 7: Final Approval Actions

Click Add New – Field Update

- Select Field To Update: Event Status
- Name: Event Status Update
- Field: Event Status
- New Value: Canceled
- Save

Click Add New – Email Alert

- Description: Alert - Cancellation Approved

- Email Template: Approval Notification to Event Owner
- Recipient Type: Select Event Owner
- From Email Address: Select Current User's Email
- Save

Figure 30: Final Approval Actions Configuration

Step 8: Final Rejection Actions

Click Add New – Email Alert

- Description: Alert - Cancellation Rejected
- Email Template: Rejection Notification to Event Owner
- Recipient Type: Select Event Owner
- From Email Address: Select Current User's Email
- Save

Click Add New – Field Update

- Select Field To Update: Event Status
- Name: Event Status To Rejected
- Field: Event Status
- New Value: Rejected
- Save

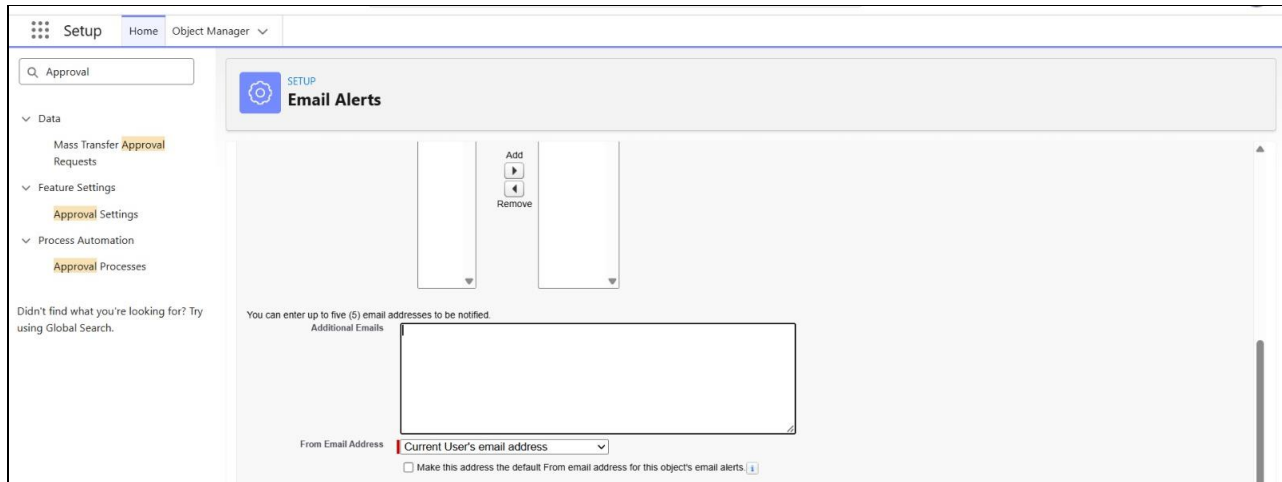


Figure 31: Final Rejection Actions Configuration

Step 9: Approval Steps

Click Add New Step

- Step Name: "Manager Review"
- Unique Name: Auto-filled
- Step Number: 2
- Click Next

Specify Step Criteria:

- Choose "All records should enter this step"
- Click Next

Select Assigned Approver:

- Select: Automatically assign to approver(s)
- Select: A specific user
- Choose: Manager (or related user)
- Click Save

Approval Processes

New Approval Step

Help for this Page

Step 1. Enter Name and Description Step 1 of 3

Next Cancel

Enter a name, description, and step number for your new approval step.

Enter Name and Description I = Required Information

Approval Process Name: Event_Cancellation_Process

Name: Manager Review

Unique Name: Manager_Review

Description:

Step Number: 2

Figure 32: Manager Review Approval Step

Step 10: Activate the Approval Process

- Click View Approval Process Detail Page
- Click the "Activate" button at the top

Step 3. Select Assigned Approver Step 3 of 3

Previous Save Cancel

Specify the user who should approve records that enter this step. Optionally, choose whether the approver's delegate is also allowed to approve these requests.

Select Approver

☐ Let the submitter choose the approver manually.

☐ Automatically assign using the user field selected earlier. (Manager)

☐ Automatically assign to queue.

☒ Automatically assign to approver(s).

User: Michael Jackson

Add Row Remove Row

When multiple approvers are selected:

☒ Approve or reject based on the FIRST response.

☐ Require UNANIMOUS approval from all selected approvers.

☐ The approver's delegate may also approve this request.

Step 3. Select Assigned Approver Step 3 of 3

Figure 33: Activated Event Cancellation Approval Process

Approval Process Testing:

A final cancellation test was performed using:

- Event: Final Cancellation Test
- Event Type: Corporate
- Event Date: 15 December 2026

- Client: Priya Sharma
- Venue: Green Garden Convention Center

The event was submitted for cancellation approval.

Approval History:

Step Name	Date	Status	Assigned To
Manager Review	10/1/2026, 11:48 PM	Approved	Raj CEO
Approval Request Submitted	10/1/2026, 11:41 PM	Submitted	Mythili S

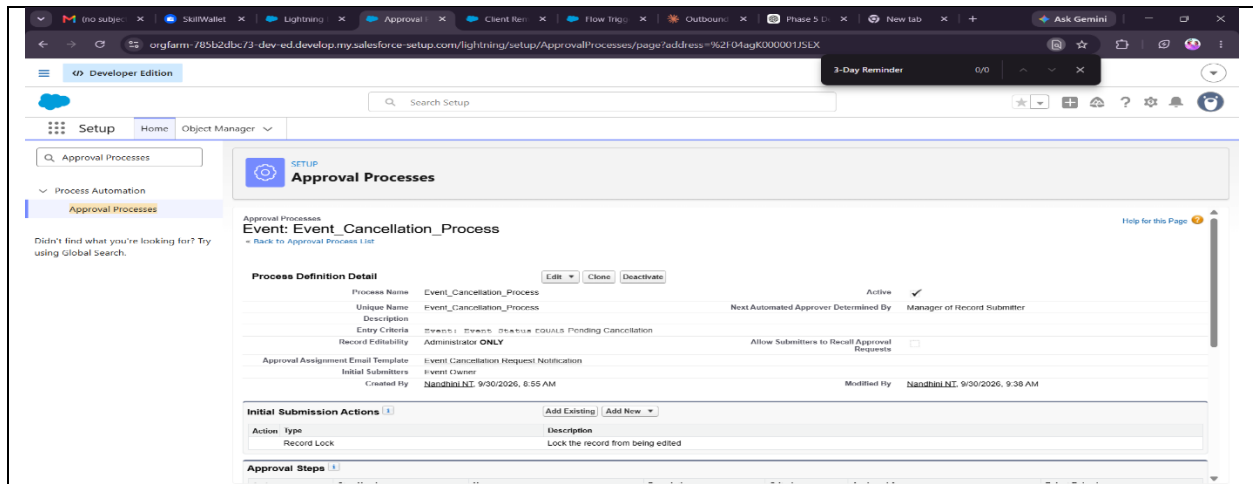


Figure 34: Successful Cancellation Approval and Approval History

After approval:

- Event Status = Canceled
- Venue = Available

6.2.5 Milestone 7: Flow – 3-Day Reminder Email

What is a Flow?

In Salesforce, a flow is a powerful tool that allows automation of business processes, collection and updating of data, and guiding users through a series of screens or steps. Flows are built using a visual interface and can be created without any coding knowledge.

Activity 1: Create Flow to Send 3-Day Reminder Email to Client

Scenario: When an Event is created/updated with Status = Confirmed and a valid Client with an email address, the system should automatically send a reminder email to the Client 3 days before the Event Date.

1. Go to Setup – type "Flow" in Quick Find box – Click on Flows and Select New Flow

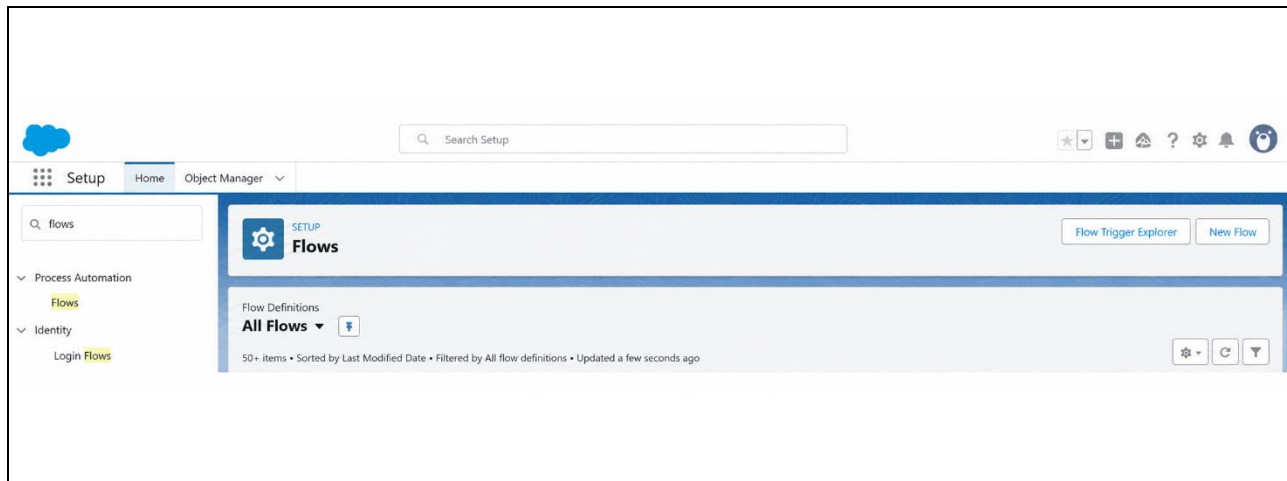


Figure 35: Flow Creation Page

2. Select Record-Triggered Flow. Click on Create
3. Under Object, select "Event"
4. Select "A record is Created or Updated"

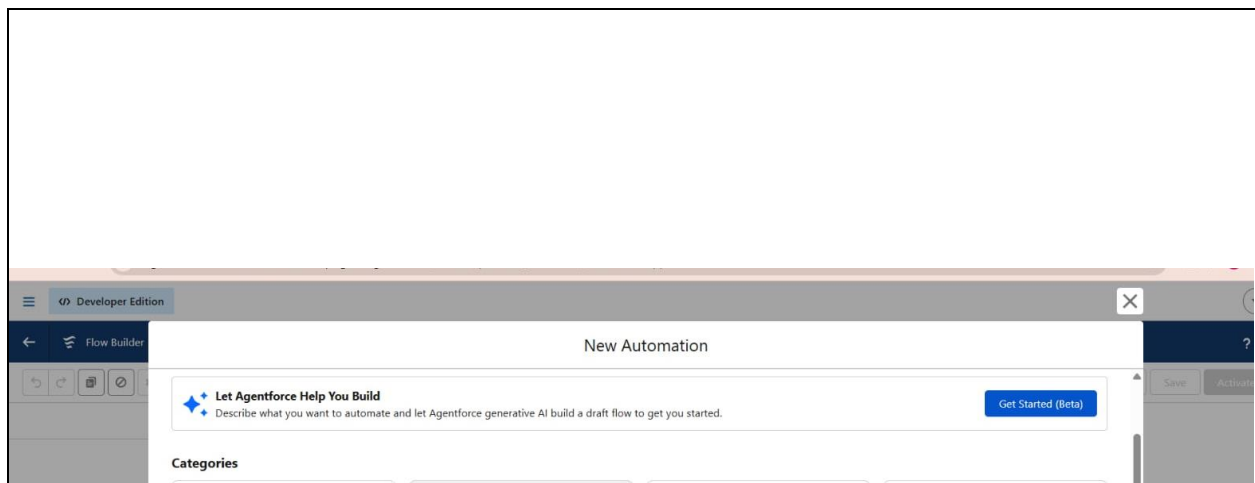


Figure 36: Flow Start Configuration

5. Enter Condition:

- Field: Event Status
 - Operator: Equals
 - Value: Confirmed
6. Select "Optimize the flow for: Actions and Related Records" and Click on Done

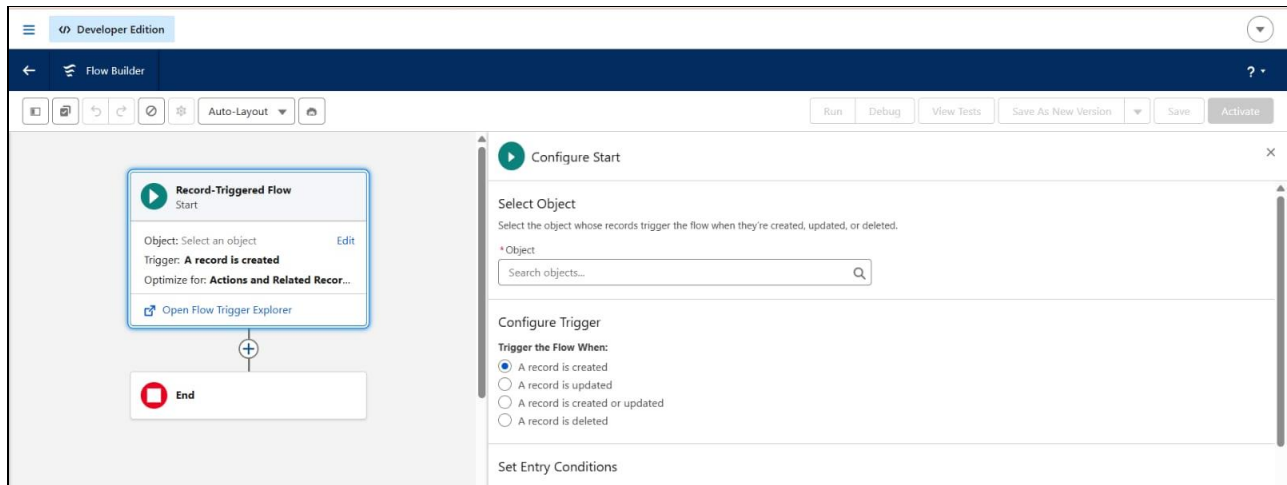


Figure 37: Flow Entry Conditions

7. Click on Start Button, and Click on Add Schedule Path

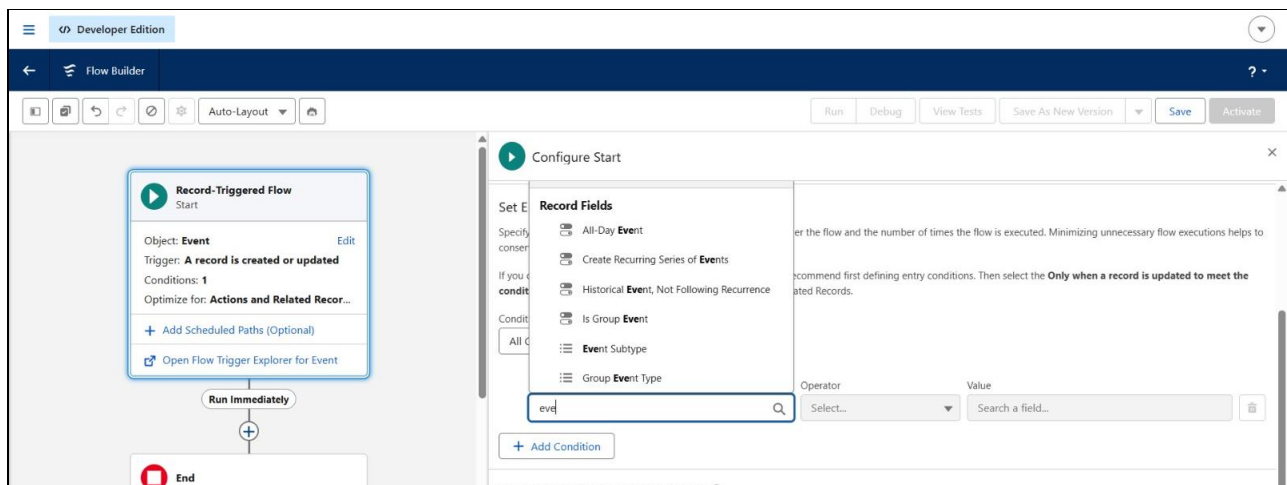


Figure 38: Scheduled Path Configuration

8. Enter Path Name as "3_Days_Before", API name automatically populates
9. Select Time Source as "Event__c: Event Date"
10. Enter Offset Number: 3
11. Select Offset Options: Days Before

Configure Scheduled Paths

SCHEDULED PATHS +

Run Immediately

3_Days_Before

SCHEDULED PATH DETAILS Delete Path

* Path Label: 3_Days_Before

* API Name ⓘ: X3_Days_Before

* Time Source: Event_c: Event Date ▼

Figure 39: Scheduled Path Details

12. Select Toolbox from Top left corner, Click on New Resource

New Resource

* Resource Type: Select... ▼

- Variable
Store a value that can be used and changed throughout the flow.
- Constant
Store a value that can be used but not changed throughout the flow.
- Formula
Calculate a value when the formula is used in the flow.
- Text Template
Store text that can be used and changed throughout the flow.
- Stage

Figure 40: New Resource Creation

13. Select Text Template

14. Enter API Name: Subject

- In the Body, enter: "Reminder: Your Event {!\$Record.Name} is in 3 days"
- Select View as Plain Text
- Click Done

Figure 41 shows the 'Edit Text Template' interface. It features a title bar 'Edit Text Template'. Below the title bar, there are three main sections: 'API Name' (with a value of 'Subject'), 'Description' (empty), and 'Body' (containing a search bar with the placeholder 'Insert a resource...', a magnifying glass icon, and a 'View as Plain Text' button with a dropdown arrow). Each section has an asterisk and an information icon to its left.

Figure 41: Email Subject Text Template

Create Another Resource:

- Select Text Template
- Enter API Name: Body
- In the Body, enter:

Hello {!\$Record.Client__r.Name},

This is a reminder that your event "{!\$Record.Name}" will take place on
{!\$Record.Event_Date__c}.

Venue: {!\$Record.Venue__r.Name}

Type: {!\$Record.Event_Type__c}

We look forward to seeing you!

- EventForce Team

- Select View as Plain Text
- Click Done

Figure 42: Email Body Text Template

15. Under the 3 Days Before, Click on "+" Symbol
16. In the Dropdown List, select "Action"
17. Select Email Action
18. Search for "Send Email" and Click on Send Email

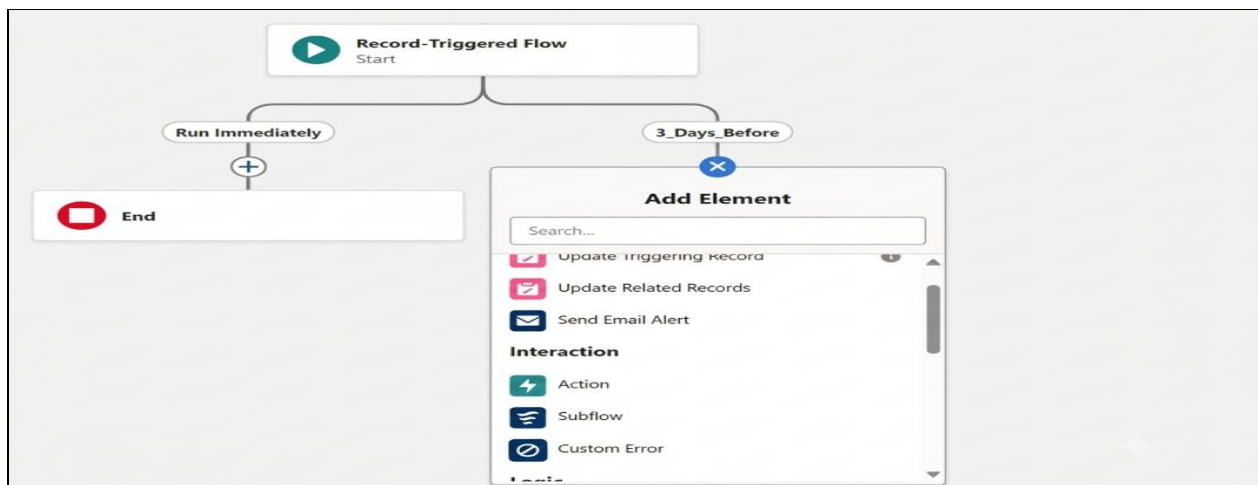
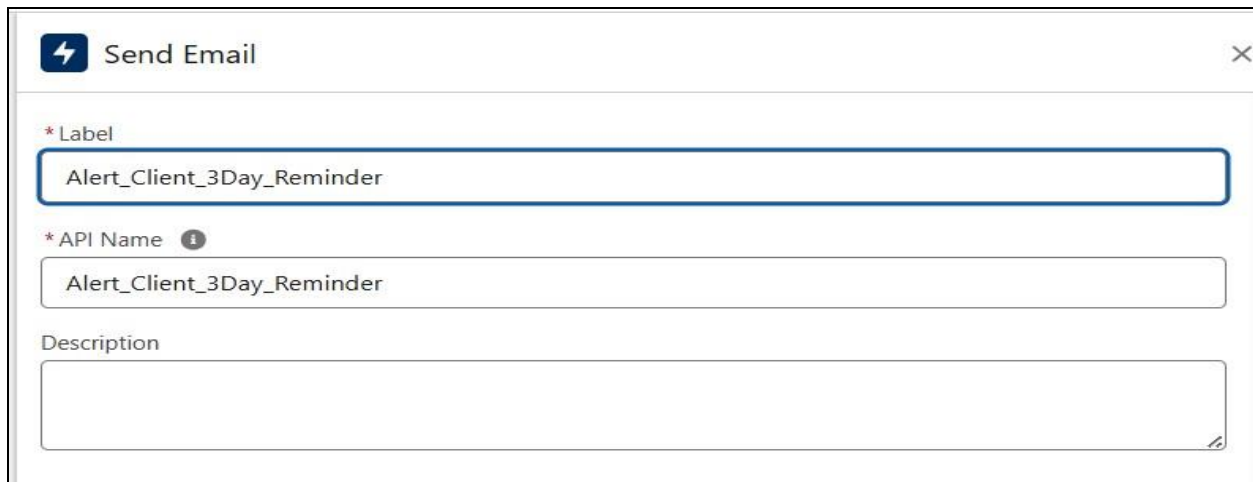


Figure 43: Send Email Action Configuration

19. Email Action: Enter the Label Name as "Alert_Client_3Day_Reminder"
20. Select the Recipient Address List as Client Email
21. Select the CC Recipient Address List: Event Owner



Send Email

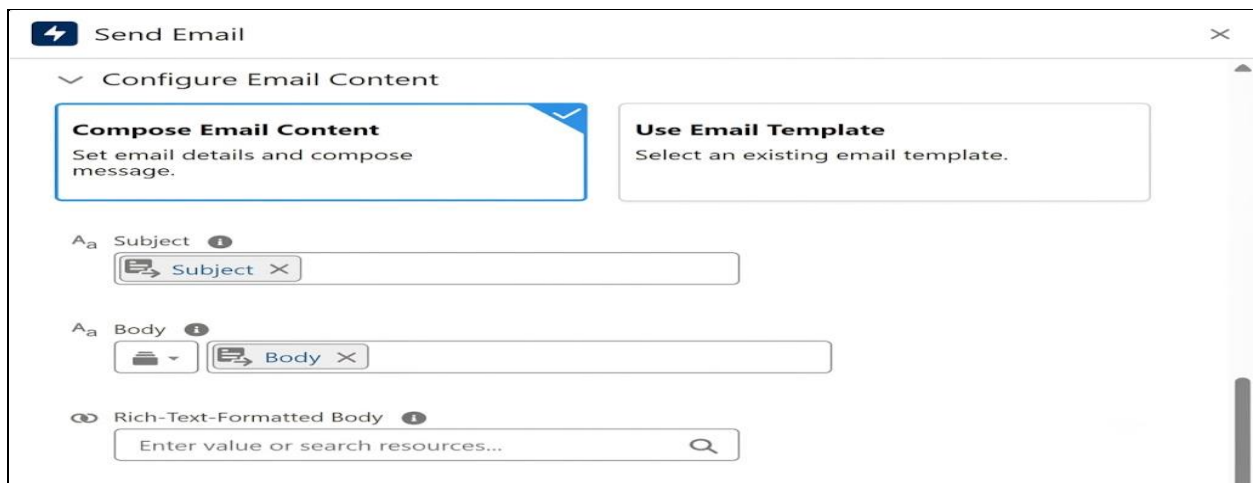
* Label
Alert_Client_3Day_Reminder

* API Name ⓘ
Alert_Client_3Day_Reminder

Description

Figure 44: Email Recipient Configuration

22. In Configure Email Content, select Compose Email Content
23. Select Subject: Subject Resource
24. Body: Body Resource
25. Click Save
26. Enter Flow Label: Client Reminder - 3 Days Before
27. Click Save & Activate



Send Email

✓ Configure Email Content

Compose Email Content
Set email details and compose message.

Use Email Template
Select an existing email template.

A_a Subject ⓘ
Subject X

A_a Body ⓘ
Body X

Rich-Text-Formatted Body ⓘ
Enter value or search resources...

Figure 46: Flow Save and Activation

6.2.6 Milestones 8-10: Apex Classes, Triggers, Batch, Scheduled Apex

Milestone 8: Apex Classes

Activity 1: Create VenueStatusHelper Class

Scenario: When an Event is created with a Venue, mark Venue Availability = "Reserved". When event is cancelled, mark "Available".

Create Apex Class:

1. Click Gear Icon and Select Developer Console
2. Click on File – New – Apex Class
3. Give name as "VenueStatusHelper" and Click Ok

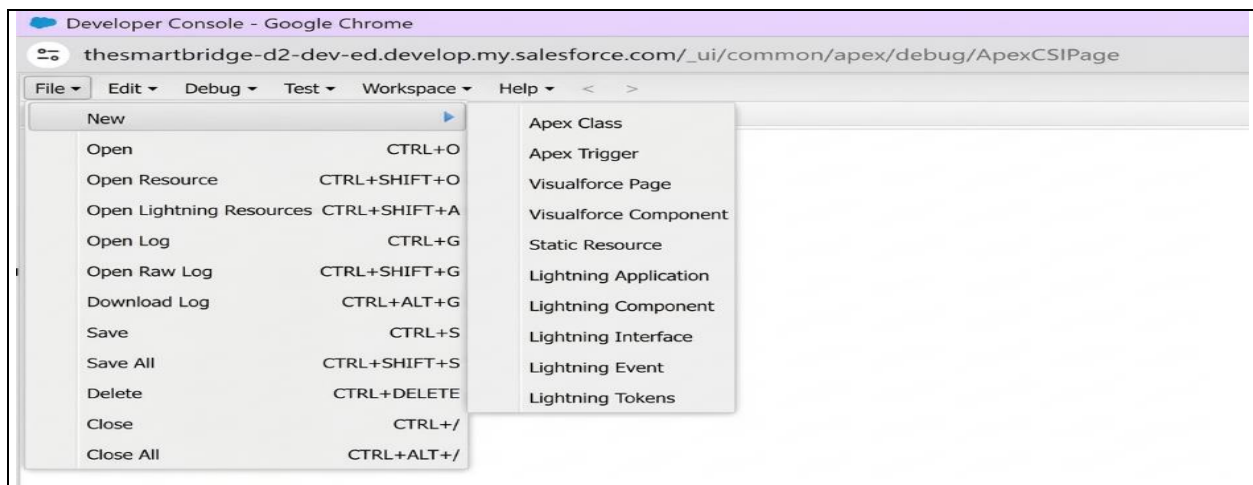


Figure 47: New Apex Class Creation

4. Write the Apex Class Code:

```
public with sharing class VenueStatusHelper {  
    public static void updateVenueStatus(List<Event__c> events){  
        List<Venue__c> venues = new List<Venue__c>();  
        for(Event__c ev : events){  
            if(ev.Venue__c != null){  
                if(ev.Event_Status__c == 'Confirmed'){  
                    venues.add(new Venue__c(Id=ev.Venue__c, Availability_Status__c='Reserved'));  
                } else if(ev.Event_Status__c == 'Canceled'){  
                    venues.add(new Venue__c(Id=ev.Venue__c, Availability_Status__c='Available'));  
                }  
            }  
        }  
        if(!venues.isEmpty()) update venues;  
    }  
}
```

5. Save the class using CTRL+S

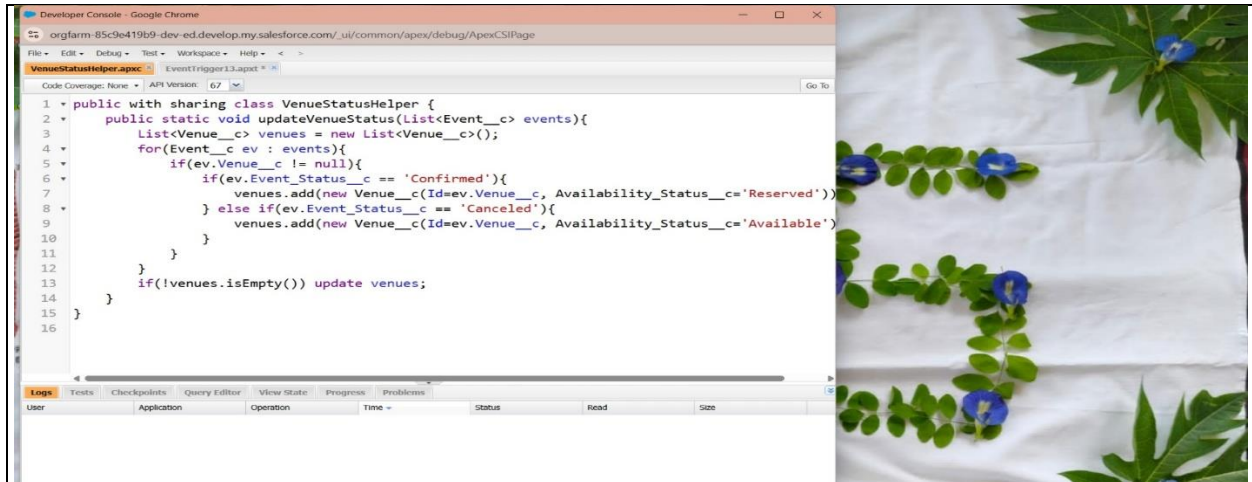


Figure 48: VenueStatusHelper Apex Class

Create Trigger for this Class:

1. Click on File – New – Apex Trigger
2. Give name as "EventTrigger13" and Select Object: Event__c
3. Click Submit



Figure 49: New Apex Trigger Creation

4. Write the Apex Trigger Code:

```

trigger EventTrigger13 on Event__c (after insert, after update) {
    VenueStatusHelper.updateVenueStatus(Triquer.new);
}

```

5. Save the trigger using CTRL+S

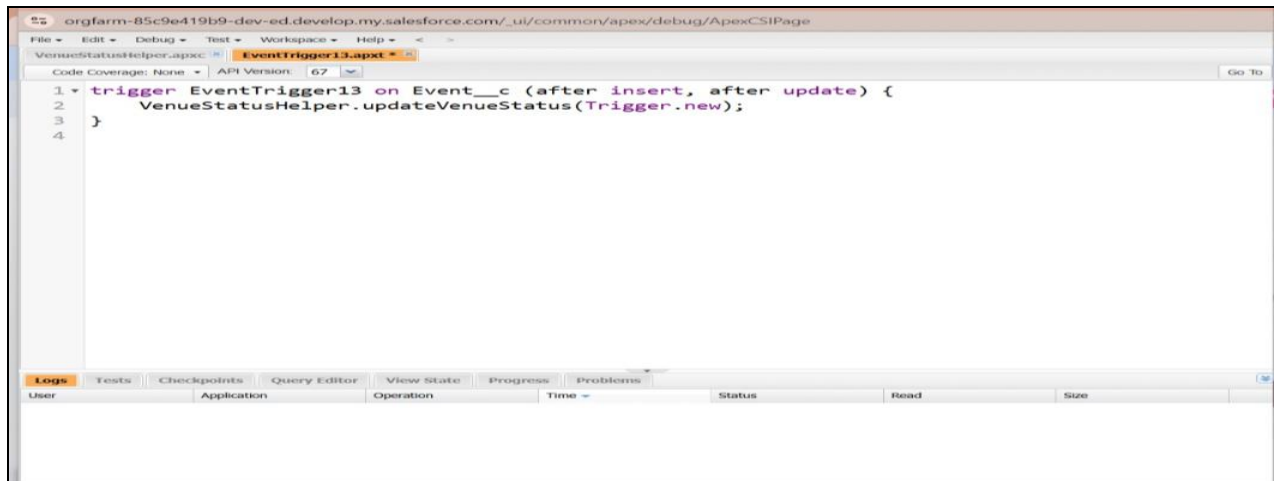


Figure 50: EventTrigger13 Apex Trigger

Milestone 9: Apex Trigger (Double Booking Prevention)

Activity 1: Create PreventDoubleBooking Trigger

Scenario: Do not allow two events at the same Venue on the same Date.

Create an Apex Trigger:

1. Click on File – New – Apex Trigger
2. Give name as "PreventDoubleBooking" and Select Object: Event__c
3. Click Submit
4. Write the Apex Trigger Code:

```

trigger PreventDoubleBooking on Event__c (before insert, before update) {
    Set<Id> venueIds = new Set<Id>();
    for(Event__c ev : Triquer.new){
        if(ev.Venue__c != null) venueIds.add(ev.Venue__c);
    }
}

```

```

Map<Id, List<Event__c>> venueEventMap = new Map<Id, List<Event__c>>();
for(Event__c e : [SELECT Id, Venue__c, Event_Date__c

```

```

        FROM Event__c WHERE Venue__c IN :venueIds]){
    if(!venueEventMap.containsKey(e.Venue__c)){
        venueEventMap.put(e.Venue__c, new List<Event__c>());
    }
    venueEventMap.get(e.Venue__c).add(e);
}

for(Event__c ev : Trigger.new){
    if(ev.Venue__c != null && venueEventMap.containsKey(ev.Venue__c)){
        for(Event__c existing : venueEventMap.get(ev.Venue__c)){
            if(existing.Event_Date__c == ev.Event_Date__c && existing.Id != ev.Id){
                ev.addError('This Venue is already booked on this date.');
```

5. Save the trigger using CTRL+S

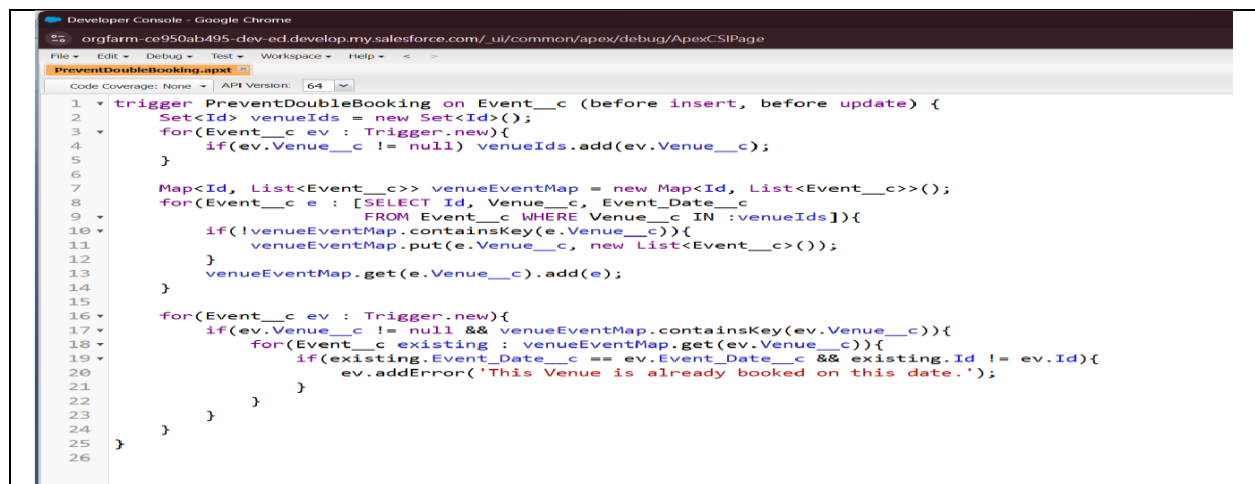


Figure 52: Double Booking Prevention Test Error

Milestone 10: Batchable and Schedulable Asynchronous Apex

Scenario: Mark all past events as Completed every night.

Step 1: Create Batch Apex Class

1. Click Gear Icon and Select Developer Console
2. Click on File – New – Apex Class
3. Give name as "BatchCompleteEvents" and Click Ok
4. Write the Apex Class Code:

```
global class BatchCompleteEvents implements Database.Batchable<sObject> {
    global Database.QueryLocator start(Database.BatchableContext bc){
        return Database.getQueryLocator(
            'SELECT Id, Event_Date__c, Event_Status__c FROM Event__c WHERE
            Event_Date__c < TODAY AND Event_Status__c != \'Completed\'"
        );
    }
    global void execute(Database.BatchableContext bc, List<Event__c> scope){
        for(Event__c ev : scope){
            ev.Event_Status__c = 'Completed';
        }
        update scope;
    }
    global void finish(Database.BatchableContext bc){
        System.debug('Past events marked as Completed');
    }
}
```

5. Save the class using CTRL+S

Step 2: Schedulable Class Implementation

1. Click Gear Icon and Select Developer Console
2. Click on File – New – Apex Class
3. Give name as "ScheduleCompleteEvents" and Click Ok
4. Write the Apex Class Code:

```
global class ScheduleCompleteEvents implements Schedulable {
    global void execute(SchedulableContext sc){
        Database.executeBatch(new BatchCompleteEvents(), 200);
    }
}
```

5. Save the class using CTRL+S

Step 3: Schedule Batch Apex from Salesforce Setup

1. Go to Setup (Click the gear icon – Setup)
2. Search for "Apex Classes" in the Quick Find box
3. Click Apex Classes
4. On the top, click the "Schedule Apex" button

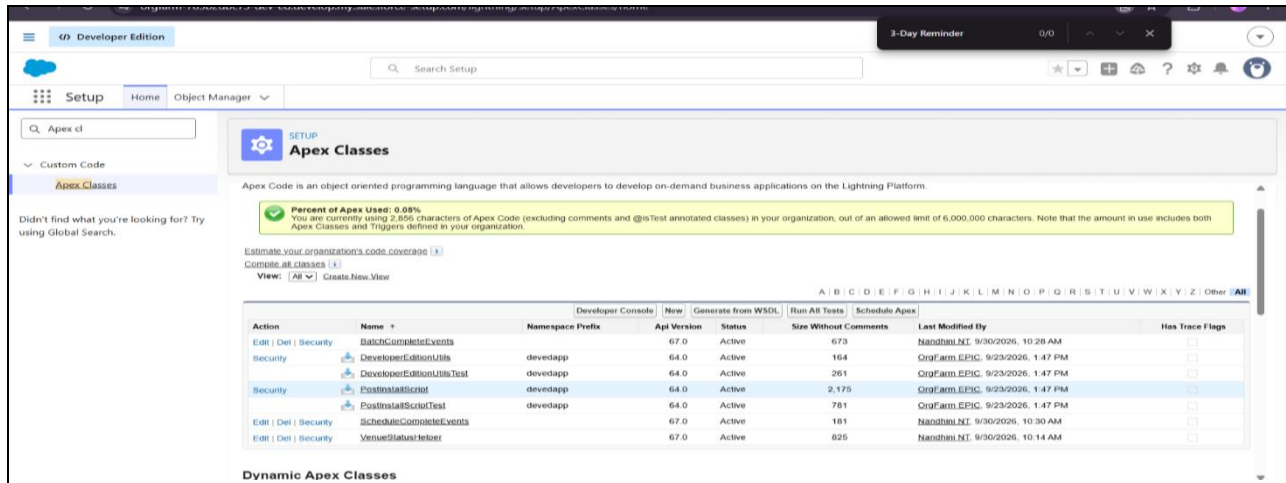


Figure 56: Schedule Apex Button

5. Enter Job Details:
 - Job Name: Daily Event Completion
 - Apex Class: Select ScheduleCompleteEvents
 - Frequency: Weekly
 - Days: Select All Days
 - Preferred Start Time: 08:00 PM
6. Click Save – Job is now scheduled

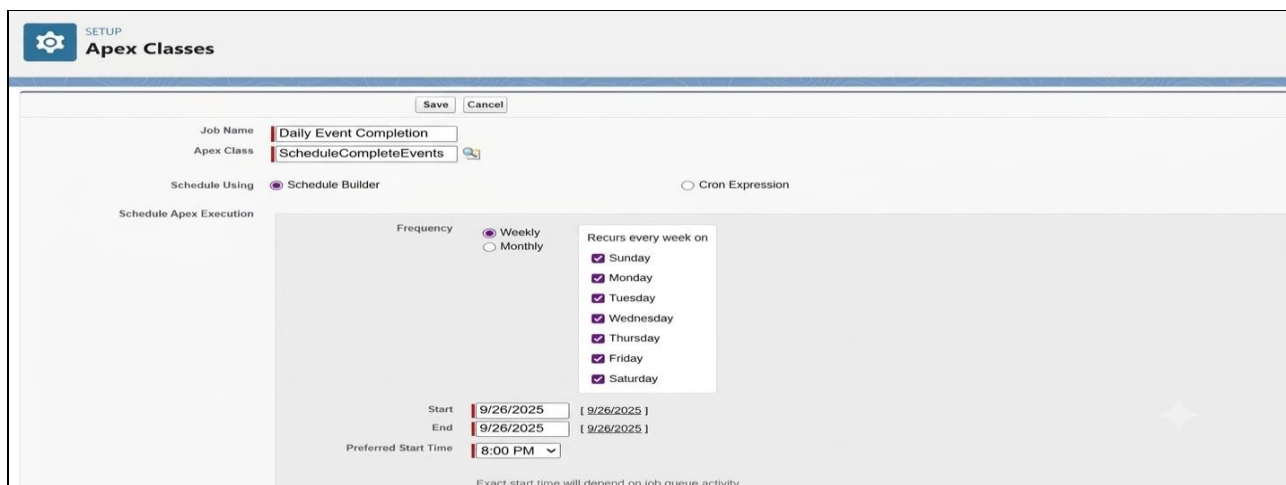


Figure 58: Past Event Automatically Marked Completed

6.3 Phase 3: UI/UX Development & Customization

UI/UX Development & Customization focuses on creating a simple, user-friendly interface for the EventForce Management System. This phase includes the creation of a custom Lightning App in Salesforce to organize and access all modules such as Events, Clients, Vendors, Venues, and Feedback from a single workspace.

Additionally, Reports and Dashboards were developed to visualize event schedules, vendor performance, and client feedback. Reports such as "Upcoming Events by Month" were used for analytical tracking, while the EventForce Operations Dashboard provides a clear visual summary of key business metrics.

This phase ensures a responsive and data-driven user experience, allowing users to manage and monitor event operations efficiently through Salesforce's Lightning interface.

6.3.1 Milestone 11: The Lightning App

What is an App?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1: Create a Lightning App

Create the Event Planner Management System App:

1. From Setup, enter "App Manager" in the Quick Find and select App Manager

New Lightning App

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

* App Name ⓘ
Event Planner

* Developer Name ⓘ
Event_Planner

Description ⓘ
Enter a description...

App Branding

Image ⓘ
[Image of Event Planner logo]

Primary Color Hex Value ⓘ
#0070D2

Clear

Org Theme Options
☐ Use the app's image and color instead of the org's custom theme

Next

Figure 59: Lightning Experience App Manager

2. Click New Lightning App
3. Enter "Event Planner" as the App Name
4. Upload Image, then click Next

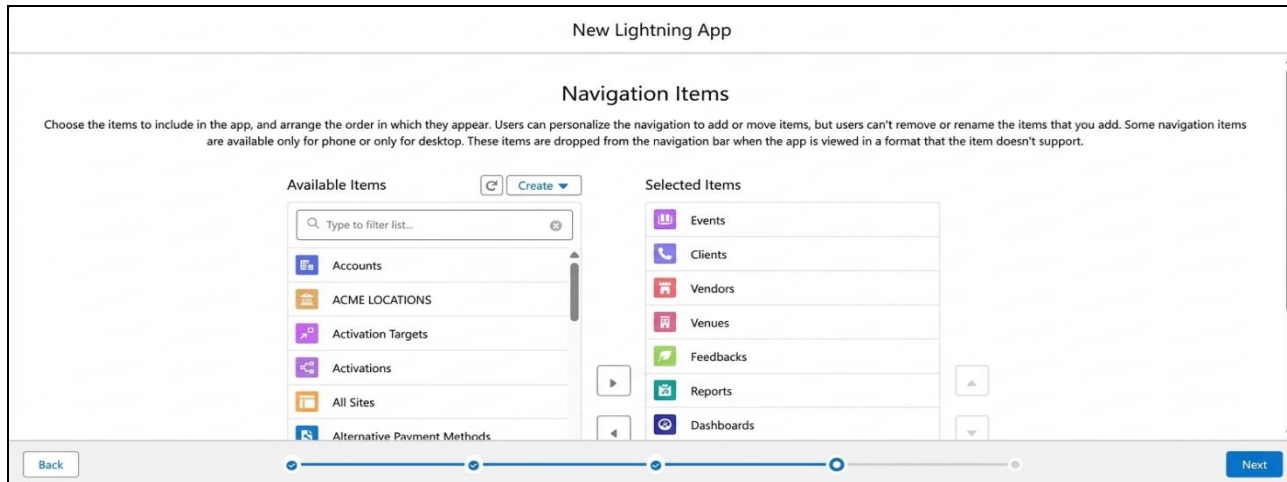


Figure 60: New Lightning App Configuration

5. Under App Options, leave the default selections and click Next
6. Under Utility Items, leave as it is and click Next
7. From Available Items, select Event, Client, Vendor, Venue, Feedback, Reports, and Dashboards and move them to Selected Items. Click Next

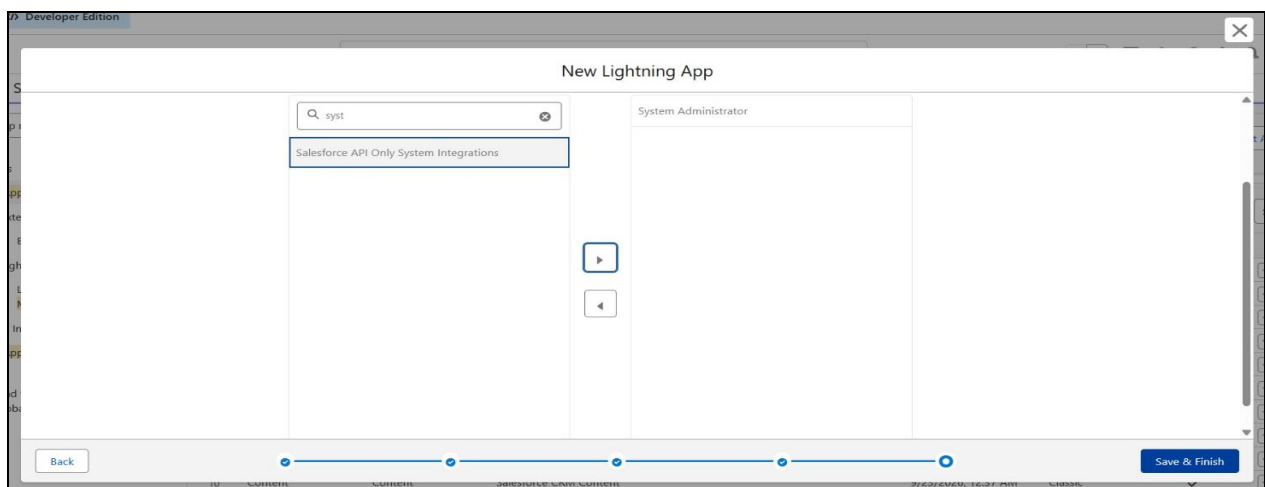


Figure 61: Navigation Items Selection

8. From Available Profiles, select System Administrator and move it to Selected Profiles
9. Click Save & Finish

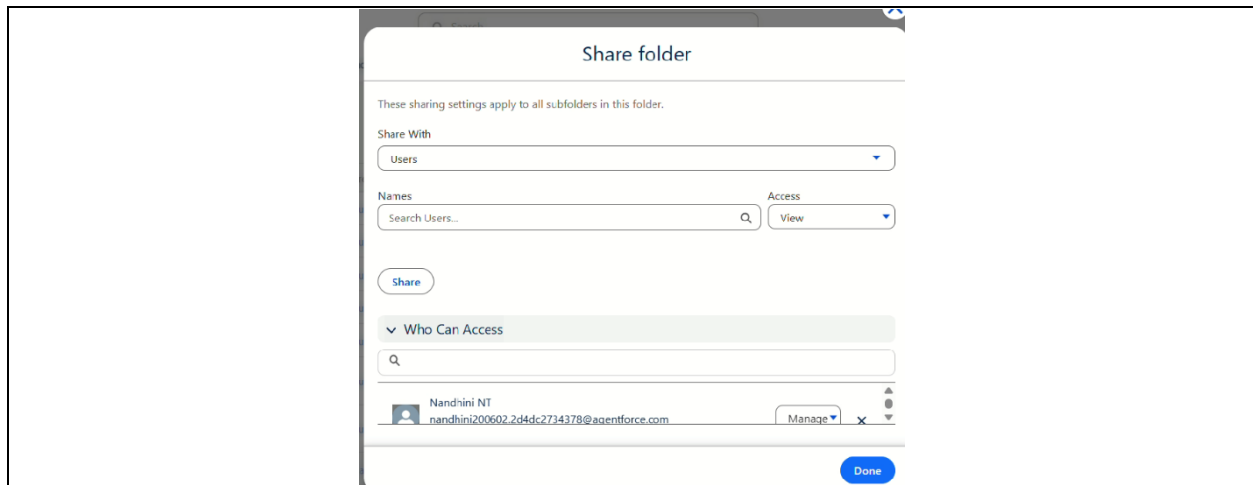


Figure 62: User Profiles Selection

To verify changes: Click the App Launcher, type "Event Planner" and select the Event Planner app.

Note:

1. App Launcher - Displays available apps
2. App Name - Displays the current selected app
3. Navigation Menu - Displays the tabs available inside the app

6.3.2 Milestones 12-13: Reports and Dashboard

Milestone 12: Reports

Activity 1: Create a Report Folder

1. Click on the App Launcher and search for "Reports"
2. Click on the Reports tab, click on New Folder
3. Enter the Folder Label as "EventForce Reports" (Folder unique name will be auto-populated)
4. Click Save

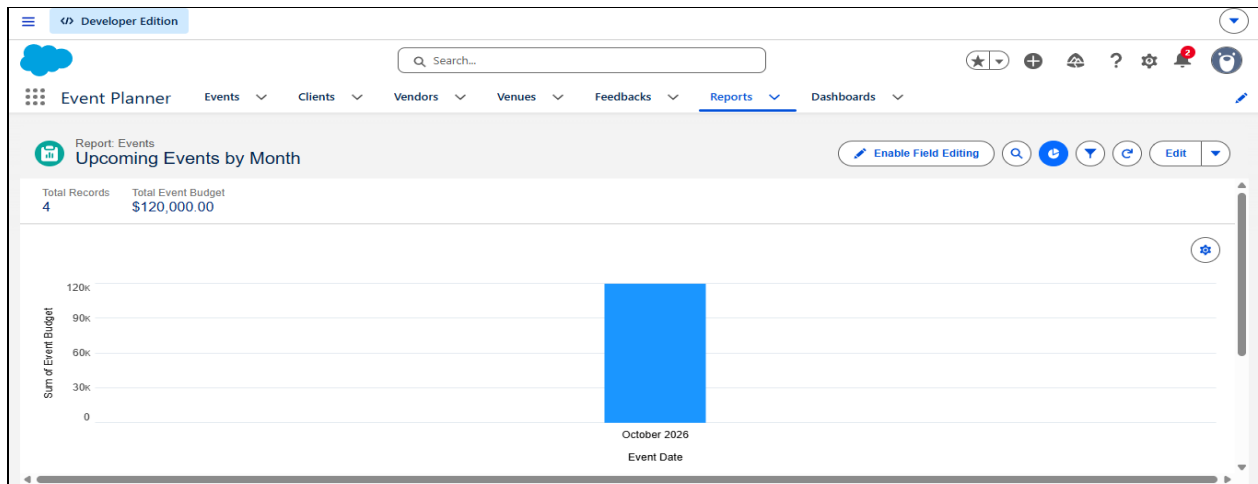


Figure 64: EventForce Reports Folder Creation

Activity 2: Sharing a Report Folder

1. Go to the App – click on the Reports tab
2. Click on the All folder, click on the Dropdown arrow for EventForce Reports Folder, and Click on Share
3. Select the share with "Roles", in the Name field search and select "Event Coordinator", "Vendor Manager" and give "View" as access for that role
4. Click Share, and click on Done

Activity 3: Create Report

Note: Before creating a report, create at least "10" records in every object. Try to fill every field in each record for better experience.

Report 1: Upcoming Events by Month

Use: See upcoming events and counts by month.

1. Go to the App Launcher – click on the Reports tab
2. Click New Report
3. Select the Category as All, search for "Events" or "Events with Client & Venue", select it, and click Start Report
4. In the outline pane, select the fields mentioned below in the Columns section:
 - Event Name
 - Event Date
 - Event Type

- Client__r.Name
 - Venue__r.Name
 - Event_Budget__c
5. Remove any unnecessary fields
 6. Toggle Update Preview Automatically
 7. Go to Filters and apply the filters as follows:
 - Show Me = All Events
 - Event Date – TODAY (or Next 30 Days)
 - Event Status – Canceled
 8. In the GROUP ROWS section, select the fields:
 - Event Date – click the drop-down – Group by Calendar Month
 9. Click on Add Summary, summarize the following:
 - Row Count (auto)
 - Event_Budget__c – SUM (to see total budget per month)
 10. Click on Add Chart, select:
 - Chart Type: Column or Line
 - X-axis = Event Date (Calendar Month)
 - Y-axis = Row Count or SUM(Event_Budget__c)
 11. Click Save, give the report name: "Upcoming Events by Month"
 12. The Report Unique Name is auto-populated
 13. Select the folder EventForce Reports – click Save – Run

Event Date	Event Budget	Event Type	Event: Event	Client	Venue
November 2026 (3)	\$10,000.00	Birthday	Birthday Party	-	-
	\$30,000.00	Corporate	Annual Meeting	-	-
	\$60,000.00	Festival	Music Festival	-	-
Subtotal	\$100,000.00				
October 2026 (5)	\$50,000.00	Wedding	Wedding Celebration	Gin	Hotel
	\$30,000.00	Corporate	Tech Conference	John	Hotel
	\$10,000.00	Birthday	Birthday Party	Rangrang	beach
	\$50,000.00	Wedding	Wedding Celebration	-	-
	\$30,000.00	Corporate	Corporate Meetup	-	-
Subtotal	\$170,000.00				
September 2026 (1)	\$40,000.00	Concert	Music Festival	Sunjay	Road
Subtotal	\$40,000.00				
Total (9)	\$310,000.00				

Figure 66: Upcoming Events by Month Report

Milestone 13: Dashboards

Activity 1: Create Dashboard

Create EventForce Dashboard:

1. Click on the Dashboards tab from the EventForce Management System application
2. Click on New Dashboard
3. Give name – "EventForce Operations Dashboard"
4. Click Create
5. Click on +Widget
6. Select the "Upcoming Events by Month" Report
 - Chart: Donut Chart
 - Max Values Displayed: 6
 - Subtitle: Upcoming Events Overview
 - Widget Theme: Light
7. Click Save

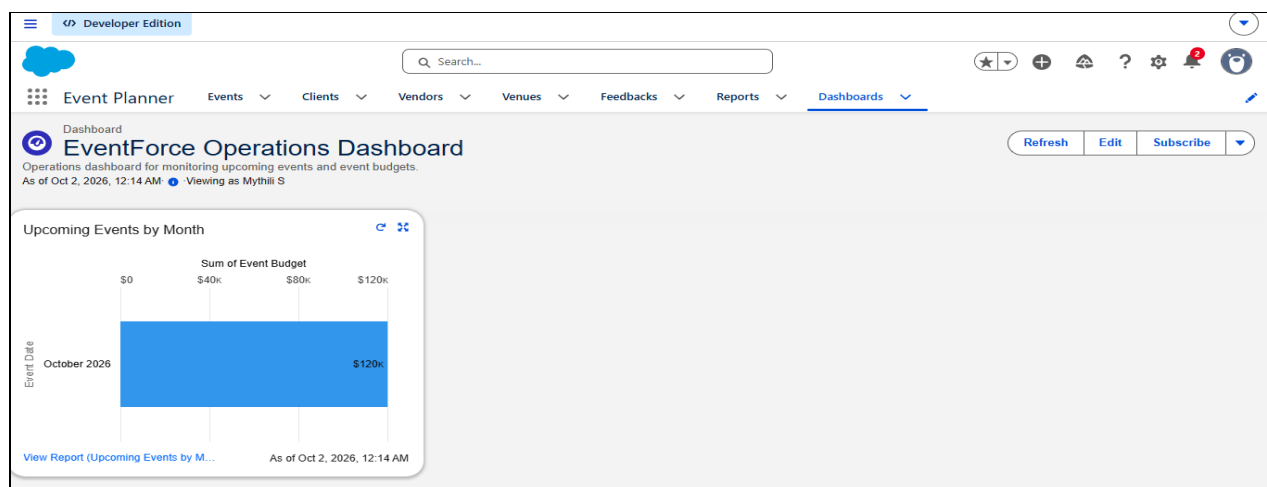


Figure 67: EventForce Operations Dashboard

6.4 Phase 4: Data Migration, Testing & Security

Data Migration, Testing & Security focuses on preparing and securing the system before deployment. This phase involved loading validated data into Salesforce, creating users, and implementing role-based access controls to ensure proper data visibility and protection.

During this phase, records for Events, Clients, Vendors, Venues, and Feedback were imported using the Data Import Wizard after pre-validation checks to avoid duplicates and maintain consistency.

User accounts were created for Event Admin, Event Coordinator, Vendor Manager, and Client, each assigned specific Profiles and Roles aligned with their access levels.

Security configurations included Profiles, Role Hierarchy, Permission Sets, and Sharing Rules to manage visibility and ensure compliance with business workflows. The Organization-Wide Defaults (OWD) were set to Private, allowing controlled record access through sharing rules like Event_Sharing_For_Vendors.

Testing activities validated data accuracy, automation performance, and user permissions.

This phase ensured that the EventForce Management System was secure, reliable, and fully functional for real-time business operations.

6.4.1 Data Migration

Sample records for Events, Clients, Vendors, Venues, and Feedback were loaded after pre-validation to avoid duplicates and keep data consistent using the Data Import Wizard and manual entry.

Sample Clients:

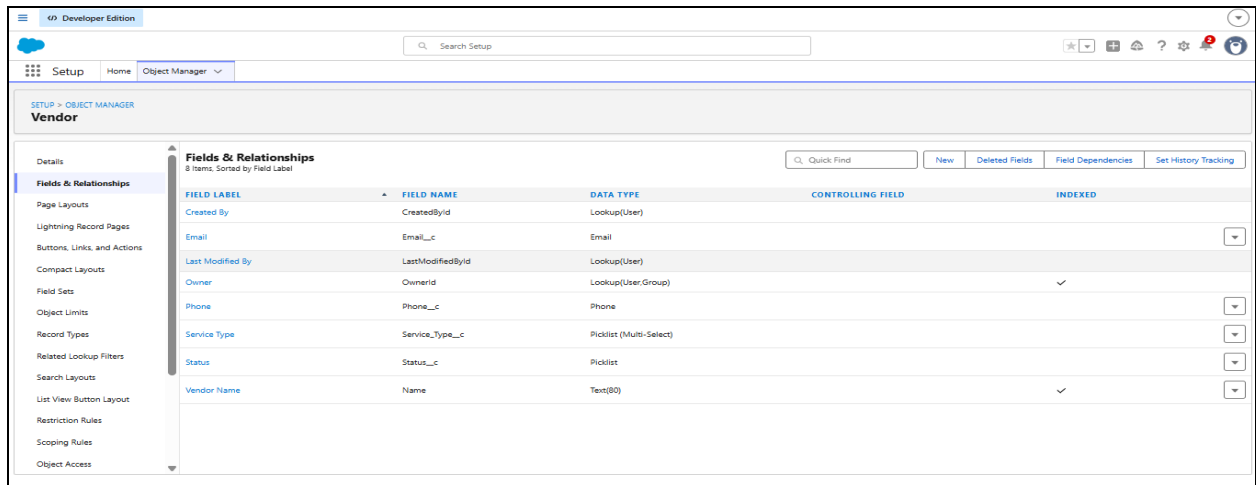
Client Name	Email	City
Arun Kumar	arun.kumar@example.com	Coimbatore
Priya Sharma	priya.sharma@example.com	Chennai
Validation Test Client	invalid-email	●

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Address	Address__c	Text Area(255)		
City	City__c	Picklist		
Client Name	Name	Text(80)		✓
Country	Country__c	Picklist		
Created By	CreatedById	Lookup(User)		
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User/Group)		✓
Phone	Phone__c	Phone		

Figure 68: Clients List View

Sample Vendors:

Vendor Name	Service Type	Status
Pixel Events	Photography	Active
Royal Catering	Catering	Active
Arun Wedding - Pixel Events	●	●



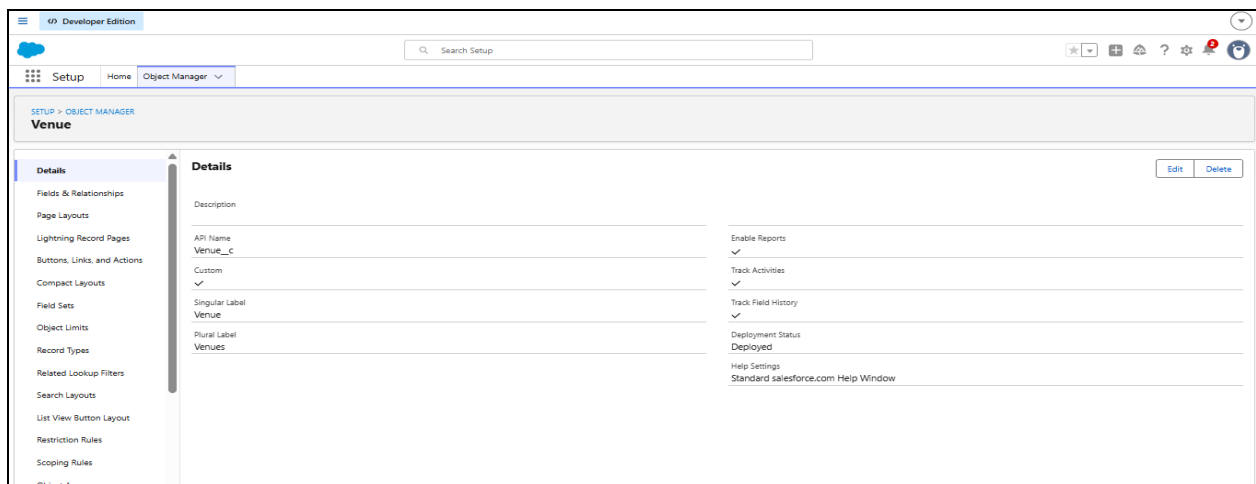
The screenshot shows the Salesforce Setup interface for the 'Vendor' object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, and others. The main content area is titled 'Fields & Relationships' and shows a list of 8 fields. The fields are sorted by Field Label. The table below represents the data shown in the screenshot.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedBy	Lookup(User)		
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User:Group)		✓
Phone	Phone__c	Phone		
Service Type	Service_Type__c	Picklist (Multi-Select)		
Status	Status__c	Picklist		
Vendor Name	Name	Text(80)		✓

Figure 69: Vendors List View

Sample Venues:

Venue Name	Capacity	Availability Status
Grand Palace Hall	500	Available/Reserved
Green Garden Convention Center	800	Available



The screenshot shows the Salesforce Setup interface for the 'Venue' object, specifically the 'Details' page. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, and others. The main content area is titled 'Details' and shows various configuration options for the Venue object. The table below represents the data shown in the screenshot.

Field Name	Value
Description	
API Name	Venue__c
Custom	✓
Singular Label	Venue
Plural Label	Venues
Enable Reports	✓
Track Activities	✓
Track Field History	✓
Deployment Status	Deployed
Help Settings	Standard salesforce.com Help Window

Figure 70: Venues List View

Sample Events:

Event Name	Event Type	Event Date	Status	Client	Venue
Arun Wedding Celebration	Wedding	15 Oct 2026	Confirmed	Arun Kumar	Grand Palace Hall
Priya Corporate Meet	Corporate	20 Oct 2026	Confirmed	Priya Sharma	Green Garden Convention Center
Arun Product Launch	Corporate	25 Oct 2026	Confirmed	Arun Kumar	Grand Palace Hall
Priya Engagement Ceremony	Wedding	5 Nov 2026	Confirmed	Priya Sharma	Green Garden Convention Center
Arun Family Reception	Wedding	12 Nov 2026	Confirmed	Arun Kumar	Grand Palace Hall
Reminder Flow Test	Corporate	●	Confirmed	●	●
Venue Automation Test	●	●	Confirmed	●	●
Final Cancellation Test	Corporate	15 Dec 2026	Canceled	Priya Sharma	Green Garden Convention Center
Past Event Completion Test	Corporate	9/30/2026	Completed	●	Green Garden Convention Center

Developer Edition

Setup Home Object Manager

Search Setup

Search: profi

Users

Profiles

Profile Event Coordinator

Users with this profile have the permissions and page layouts listed below. Administrators can change a user's profile by editing that user's personal information.

If your organization uses Record Types, use the Edit links in the Record Type Settings section below to make one or more record types available to users with this profile.

Login IP Ranges (0) | Enabled Apex Class Access (0) | Enabled Visualforce Page Access (0) | Enabled External Data Source Access (0) | Enabled Named Credential Access (0) | Enabled External Credential Principal Access (0) | Enabled Custom Metadata Type Access (0) | Enabled Custom Setting Definitions Access (0) | Enabled Flow Access (0) | Enabled Service Presence Status Access (0) | Enabled Custom Permissions (0)

Profile Detail

Name

Event Coordinator

User License

Salesforce Platform

Description

Created By

Nandhini.NT, 9/30/2026, 11:40 PM

Custom Profile

Modified By

Nandhini.NT, 9/30/2026, 11:44 PM

Figure 71: Events List View

Sample Feedbacks:

Feedback Name	Rating	Comments
Arun Wedding Feedback	5	Excellent event management and service
Priya Corporate Feedback	4	Well organized and professionally managed
Arun Product Launch Feedback	5	Excellent coordination, venue arrangement, and vendor service
Lookup Filter Test	●	●

The screenshot shows the Salesforce Setup interface in Developer Edition. The 'Object Manager' is open for the 'Feedback' object. The 'Fields & Relationships' tab is selected, showing a list of 9 fields. The fields are sorted by label. The table includes columns for Field Label, Field Name, Data Type, Controlling Field, and Indexed status. Fields include Comments, Created By, Event, Feedback Client, Feedback Name, Last Modified By, Owner, and Rating.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Comments	Comments__c	Long Text Area(32768)		
Created By	CreatedById	Lookup(User)		
Event	Event__c	Lookup(Event)		✓
Feedback Client	Feedback_Client__c	Lookup(Client)		✓
Feedback Name	Name	Text(80)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User Group)		✓
Rating	Rating__c	Picklist		

Figure 72: Feedbacks List View

6.4.2 Milestones 14-17: Profiles, Roles, Users, Permission Sets

Milestone 14: Profile

What is a Profile?

A profile is a group/collection of settings and permissions that define what a user can do in Salesforce. A profile controls:

- Object permissions
- Field permissions
- User permissions
- Tab settings
- App settings

- Apex class access
- Visualforce page access
- Page layouts
- Record Types ,Login hours & Login IP ranges

A profile can be assigned to many users, but users can be assigned a single profile at a time.

Activity 1: Create a Custom Profile (Event Admin)

1. From Setup, enter "Profiles" in Quick Find box
2. Click On New Profile
3. Select Existing Profile as "System Administrator"
4. Enter Profile Name as "Event Admin"
5. Click On Save

Why use System Administrator as Base?

- It already has Modify All Data, Customize Application, and full CRUD access on all objects
- By cloning it, the Event Admin profile automatically inherits full access
- Permissions can then be restricted/removed if needed (but usually Event Admin = full access)

Activity 2: Create a Custom Profile (Event Coordinator)

1. From Setup, enter "Profiles" in Quick Find box
2. Click On New Profile
3. Select Existing Profile as "Standard Platform User"
4. Enter Profile Name as "Event Coordinator"
5. Click On Save
6. Click On Edit

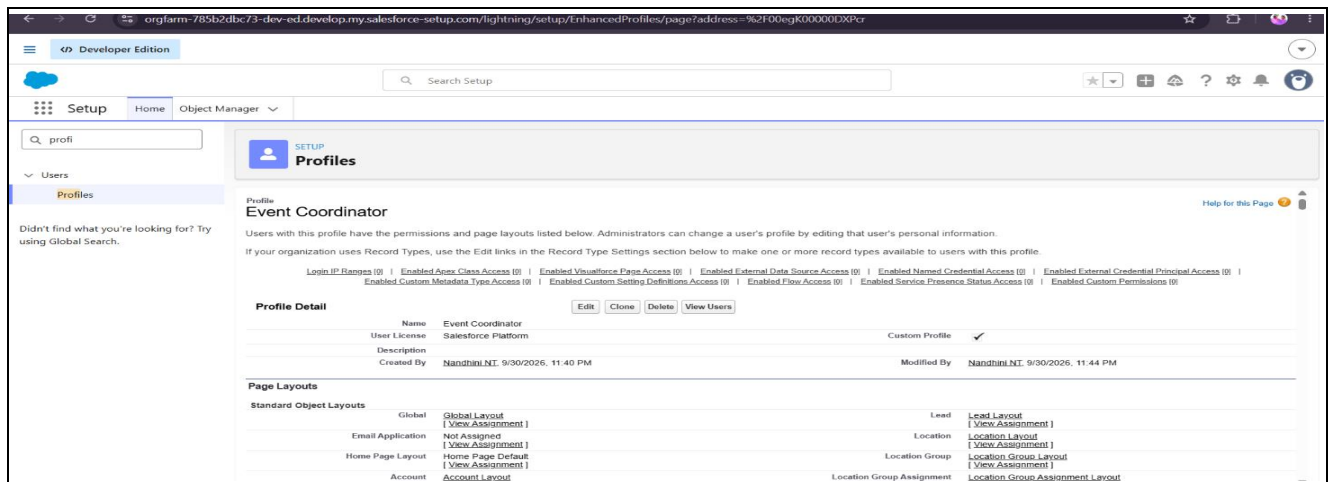


Figure 74: Event Coordinator Profile

7. Select the Custom App settings as default for the Event Planner
8. Scroll down to Custom Object Permissions Give Access:
 - Event: CRED (Create, Read, Edit, Delete)
 - Client: Read / Edit
 - Vendor: Read
 - Venue: Read
 - Feedback: Create / Read

Custom App Settings					
	Visible	Default		Visible	Default
Analytics Studio (standard__Insights)	☐	☐	Platform (standard__Platform)	☒	☐
App Launcher (standard__AppLauncher)	☐	☐	WDC (standard__Work)	☐	☐
Event Planner (Event_Planner)	☒	☒			

Figure 76: Custom Object Permissions

9. Changing the session times out after should be "2 hours of inactivity"
10. Change the password policies:

- User passwords expire in should be "never expires"
- Minimum password length should be "8"

11. Click Save

Custom Object Permissions									
	Basic Access				Data Administration				
	Read	Create	Edit	Delete	View All Records 1	Modify All Records 1	View All Fields		
ACME LOCATIONS	☐	☐	☐	☐	☐	☐	☐		
Clients	☒	☐	☒	☐	☐	☐	☐		
Events	☒	☒	☒	☒	☐	☐	☐		
Event Vendors	☐	☐	☐	☐	☐	☐	☐		
Feedbacks	☒	☒	☐	☐	☐	☐	☐		
Vendors	☒	☐	☐	☐	☐	☐	☐		
Venues	☒	☐	☐	☐	☐	☐	☐		

Figure 77: Session Settings and Password Policies

Activity 3: Create a Custom Profile (Vendor Manager)

1. From Setup, enter "Profiles" in Quick Find box
2. Click On New Profile
3. Select Existing Profile as "Standard Platform User"
4. Enter Profile Name as "Vendor Manager"
5. Click On Save
6. Click On Edit
7. Select the Custom App settings as default for the Event Planner
8. Scroll down to Custom Object Permissions Give Access:
 - Vendor: CRED
 - Event: Read Only
 - Venue: Read Only
 - Client: Read Only
 - Feedback: Read Only
9. Configure session and password settings as above
10. Click Save

Session Settings

Session Times Out After: 2 hours of inactivity

Session Security Level Required at Login: --None--

Password Policies

User passwords expire in: Never expires

Enforce password history: 3 passwords remembered

Minimum password length: 8

Password complexity requirement: Must include alpha and numeric characters

Password question requirement: Cannot contain password

Maximum invalid login attempts: 10

Lockout effective period: 15 minutes

Obscure secret answer for password resets: ☐

Require a minimum 1 day password lifetime: ☐

Don't immediately expire links in forgot password emails: ☐

Save Save & New Cancel

Figure 78: Vendor Manager Profile

Activity 4: Create a Custom Profile (Client Profile)

1. From Setup, enter "Profiles" in Quick Find box
2. Click On New Profile
3. Select Existing Profile as "Standard Platform User"
4. Enter Profile Name as "Client Profile"
5. Click On Save
6. Click On Edit
7. Select the Custom App settings as default for the Event Planner
8. Scroll down to Custom Object Permissions Give Access:
 - Event: Read Only (for their own events)
 - Feedback: Create / Read (give feedback)
 - Vendor/Venue: No Access
9. Configure session and password settings as above
10. Click Save

Custom Object Permissions								
	Basic Access				Data Administration			
	Read	Create	Edit	Delete	View All Records i	Modify All Records i	View All Fields	
ACME LOCATIONS	☐	☐	☐	☐	☐	☐	☐	
Clients	☐	☐	☐	☐	☐	☐	☐	
Events	☒	☐	☐	☐	☐	☐	☐	
Event Vendors	☐	☐	☐	☐	☐	☐	☐	

	Basic Access				Data Administration			
	Read	Create	Edit	Delete	View All Records i	Modify All Records i	View All Fields	
Feedbacks	☒	☒	☐	☐	☐	☐	☐	
Vendors	☐	☐	☐	☐	☐	☐	☐	
Venues	☐	☐	☐	☐	☐	☐	☐	

Figure 79: Client Profile

Milestone 15: Role

What are Roles?

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

Activity 1: Creation of Event Admin Role

1. From the Quick Find box, search for "Roles" and click on the Roles option
2. Select the Set Up Roles option

The screenshot shows the Salesforce Setup interface with the 'Roles' section selected. The page title is 'Creating the Role Hierarchy'. Below the title, it states: 'You can build on the existing role hierarchy shown on this page. To insert a new role, click **Add Role**.' The main content area displays 'Your Organization's Role Hierarchy' for 'VIVEKANANDHA COLLEGE OF TECHNOLOGY FOR WOMEN'. The hierarchy is shown as a tree structure with roles and their associated permissions (Edit, Del, Assign). The roles listed are: CEO, CFO, COO, SVP, Customer Service & Support, Customer Support International, Customer Support North America, Installation & Repair Services, and SVP Human Resources. Each role has an 'Add Role' link next to it. The left sidebar shows the navigation menu with 'Roles' selected under 'Setup'. The top navigation bar includes 'Setup', 'Home', and 'Object Manager'.

Figure 80: Understanding Roles Page

3. Click on Expand All and click on Add Role under CEO
4. Give Label as "Event Admin" and the Role Name gets auto-populated
5. Click on Save

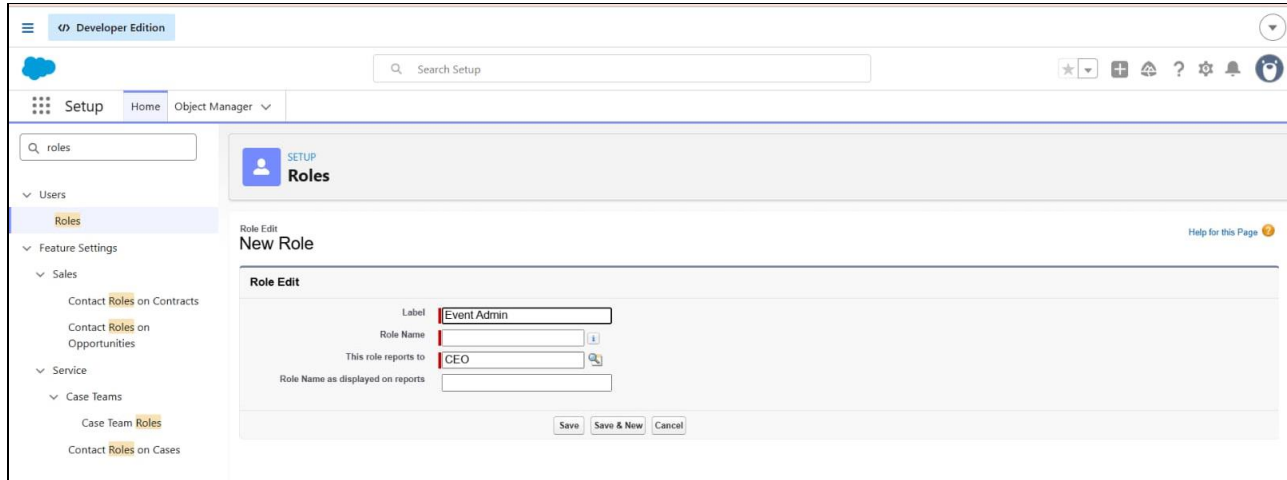


Figure 81: Event Admin Role Creation

Activity 2-4: Creating Other Roles Under Event Admin Role

Create the following roles:

1. **Event Coordinator**
 - Below Event Admin, click on Add Role
 - Enter Label as "Event Coordinator"
 - Role Name auto-populates Click Save
2. **Vendor Manager**
 - Below Event Admin, click on Add Role
 - Enter Label as "Vendor Manager"
 - Role Name auto-populates
 - Click Save
3. **Client**
 - Below Event Admin, click on Add Role
 - Enter Label as "Client"
 - Role Name auto-populates
 - Click Save

Milestone 16: User

What is a User?

A user is anyone who logs in to Salesforce. Users are employees at a company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

Note: As Salesforce license can only be used by 2 Users at a time in Dev Org, if a Salesforce license is not available, deactivate a user who has a Salesforce license or change the license type from Salesforce to any other.

Activity 1: To Create a User

1. From Setup, enter "Users" in the Quick Find box, then select Users
2. Click New User

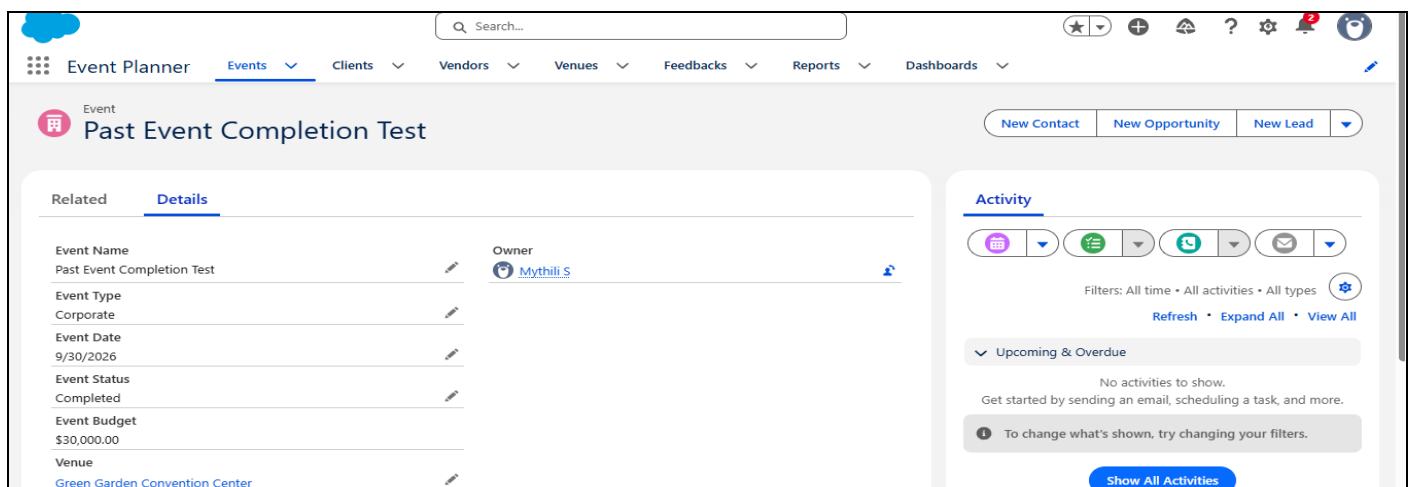


Figure 83: All Users Page

3. Fill in the fields:
 - First Name: [Name]
 - Last Name: [Name]
 - Alias: [Alias Name]
 - Email ID: [Personal Email ID]
 - Username: Username should be in this form: text@text.text and it must be Unique
 - Nick Name: [Nickname]
 - Role: Event Admin
 - User Licence: Salesforce
 - Profile: Event Admin
4. Click On Save

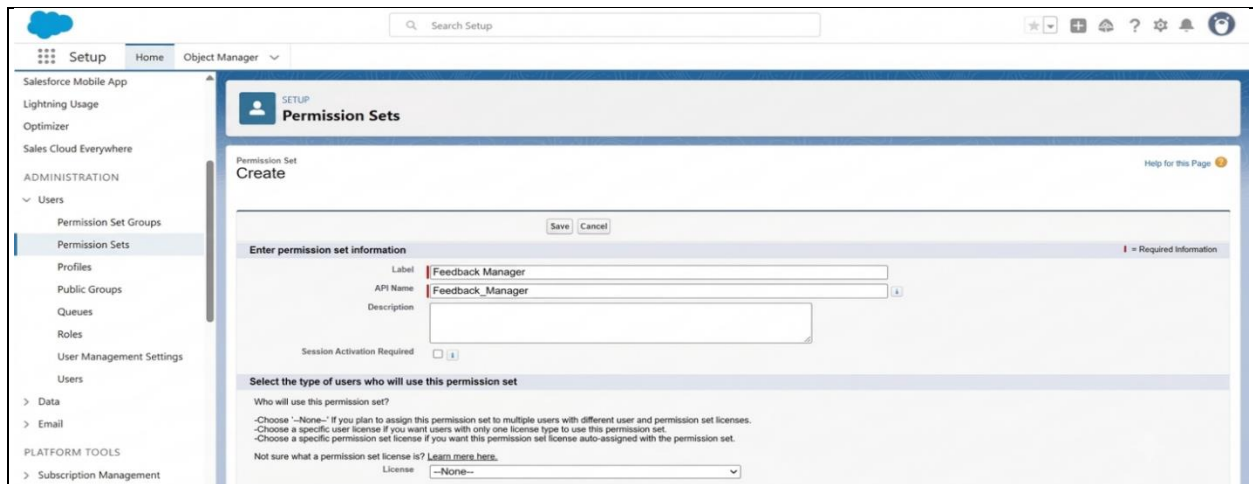


Figure 84: New User Creation

Activity 2: Creating Other Users

Repeat the steps and create other users using:

Role	User Licence	Profile
Event Coordinator	Salesforce Platform	Event Coordinator
Vendor Manager	Salesforce Platform	Vendor Manager
Client	Salesforce Platform	Client Profile

Milestone 17: Permission Set

What is a Permission Set?

A Permission Set in Salesforce is a collection of settings and permissions that grant access to various tools, features, and objects without changing the user's profile.

Activity 1: Creating New Permission Set

1. Go to Setup – type "Permission Set" in Quick Find box – Select Permission Set – click New
2. Give the Label as "Feedback Manager"
3. Click Save

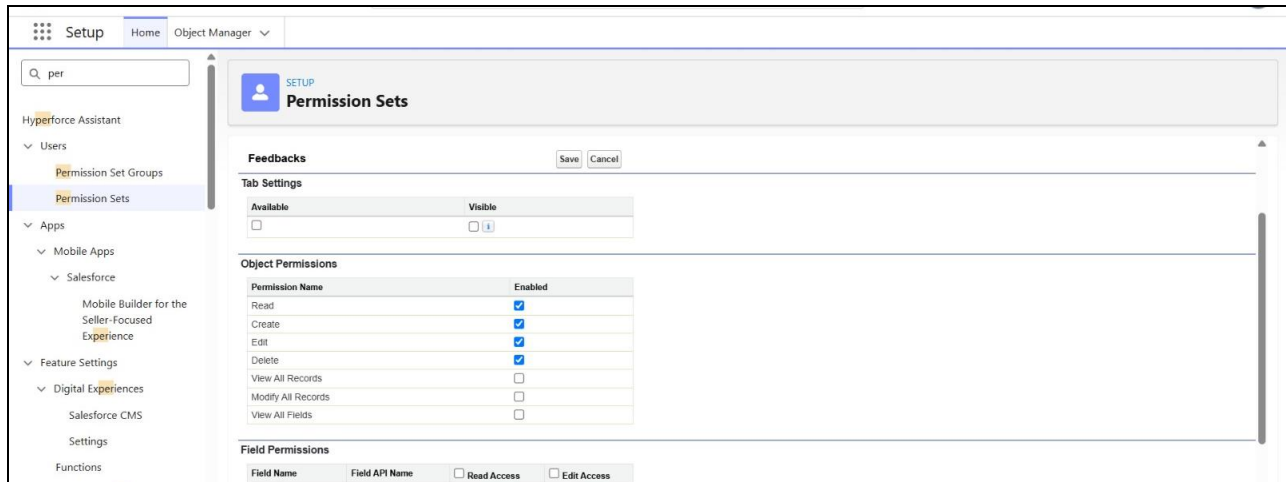


Figure 85: New Permission Set Creation

4. Select Extra Permission For Feedback Manager Permission Set
5. Go To the Object Settings
6. Search Feedback Object
7. Click On Edit, Go to the Object Permission
8. Enable The R/E/C/D Permissions
9. Click Save

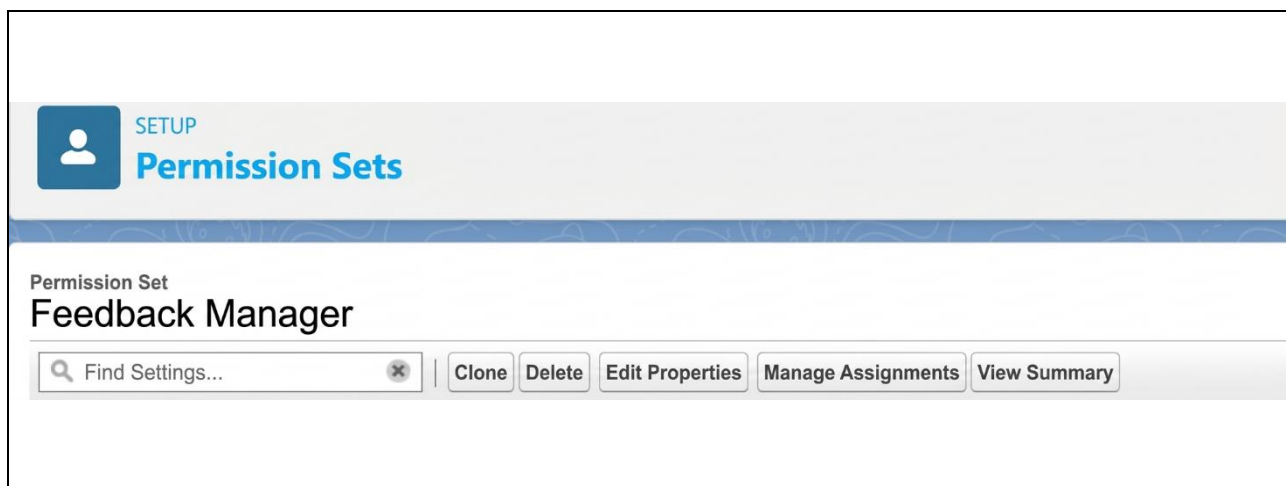


Figure 86: Feedback Manager Permission Set Configuration

10. Click On the Manage Assignments
11. Click On the Add Assignment
12. Select Event Coordinator Role User
13. Click Next & Assign

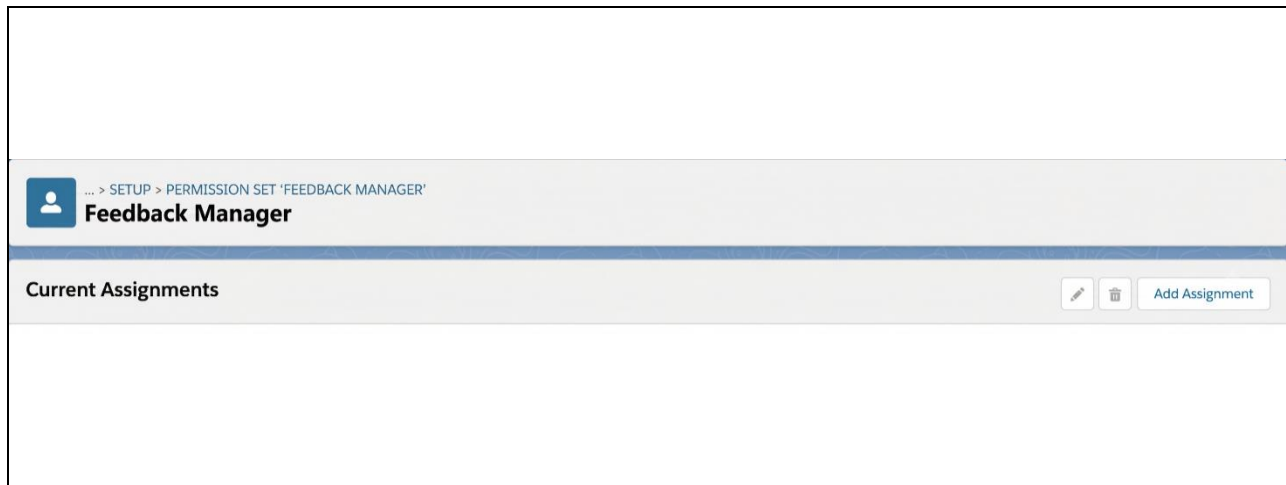


Figure 87: Permission Set Assignment

6.4.3 Milestone 18: Sharing Settings

Organization-Wide Default (OWD) Settings:

In Salesforce, OWD is the baseline level of access to records that determines what all users can see and do with records in an organization. It is used to control visibility and access for all records in Salesforce, and can be customized based on different objects.

OWD Options:

1. **Private:** Only the record owner and users above them in the role hierarchy can see or edit the record
2. **Public Read Only:** All users can view the records, but only the record owner and those above them in the role hierarchy can edit the record
3. **Public Read/Write:** All users can view and edit records, regardless of ownership
4. **Controlled by Parent:** Access level to the child record is controlled by the parent record's sharing settings

Activity 1: Configure OWD

1. Go to Setup – Sharing Settings – Click On Edit

Figure 88: Sharing Settings Page

2. Open The Sharing Setting, In Organization Wide Defaults, Select the Event Object

3. Click On Edit Button, Change the Default Internal Access as "Private", and Default External Access also as "Private" in Event Object
4. Click Save

Figure 89: Organization-Wide Defaults Configuration

Sharing Rules:

What are Sharing Rules?

Sharing rules help users share records based on conditions. They are created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend access, not restrict it.

Types of Sharing Rules:

- Owner-based Sharing Rules
- Criteria-based Sharing Rules

Activity 2: Create a Sharing Rule

1. Go to Sharing Settings, which can be found under the Quick Find section
2. Scroll down and find the Event object where a sharing rule needs to be added, and then click on New to create a new sharing rule
3. Give the Label name as "Event_Sharing_For_Vendors"
4. Rule name is auto-populated
5. In Step 3: Select which records to be shared, members of "Roles" >> "Event Coordinator"

The screenshot shows the Salesforce Setup interface for Sharing Settings. The left sidebar contains a search bar with 'sharing' and a list of navigation items: Security, Guest User Sharing Rule Access Report, and Sharing Settings (highlighted). The main content area is titled 'Sharing Settings' and includes a sub-header 'Organization-Wide Defaults' with an 'Edit' button. Below this is a table with the following data:

Object	Default Internal Access	Default External Access	Grant Access Using Hierarchies
Lead	Public Read/Write/Transfer	Private	✓
Account and Contract	Public Read/Write	Private	✓
Contact	Controlled by Parent	Controlled by Parent	✓
Order	Controlled by Parent	Controlled by Parent	✓

Figure 90: Sharing Rule - Select Records to Share

6. In Step 4: Share with, select "Roles" >> "Vendor Manager"

Object	Public Read/Write	Private	Share
Tableau Host Mapping	Public Read Only	Private	☒
User Presence	Public Read Only	Private	☒
Waitlist	Private	Private	☒
Web Cart Document	Private	Private	☒
Work Order	Private	Private	☒
Work Plan	Private	Private	☒
Work Plan Template	Private	Private	☒
Work Step Template	Private	Private	☒
Work Type	Private	Private	☒
Work Type Group	Public Read/Write	Private	☒
ACME LOCATION	Public Read/Write	Private	☒
Client	Public Read/Write	Private	☒
Event	Private	Private	☒
Feedback	Public Read/Write	Private	☒
Vendor	Public Read/Write	Private	☒

Figure 91: Sharing Rule - Select Users to Share With

7. In Step 5: Change the access level to "Read Only"

Object	New	Recalculate	Help
Work Type Group Sharing Rules	New	Recalculate	Work Type Group Sharing Rules Help
ACME LOCATION Sharing Rules	New	Recalculate	ACME LOCATION Sharing Rules Help
Client Sharing Rules	New	Recalculate	Client Sharing Rules Help
Event Sharing Rules	New	Recalculate	Event Sharing Rules Help
Feedback Sharing Rules	New	Recalculate	Feedback Sharing Rules Help

Figure 92: Sharing Rule - Select Access Level

8. Click on Save

7. FUNCTIONAL & PERFORMANCE TESTING

7.1 Testing Approach

Testing was conducted throughout the development lifecycle using a combination of manual testing and automated validation.

Testing Objectives:

1. Verify that all functional requirements are met
2. Validate that automation works as designed
3. Confirm that security controls are effective
4. Ensure data integrity across all objects
5. Verify performance under expected load

Testing Types:

Testing Type	Description	Tools/Methods
Unit Testing	Test individual components	Developer Console, Manual
Integration Testing	Test interactions between components	Manual, End-to-end scenarios
Functional Testing	Test business requirements	Manual, Test cases
Security Testing	Test access controls	Profile-based testing
Performance Testing	Test system responsiveness	Manual observation
Regression Testing	Re-test after changes	Manual, Repeat test cases

Test Environment:

- Salesforce Developer Edition
- Lightning Experience
- Sample data loaded (10+ records per object)
- Test users with different profiles
- Dedicated test records for each feature

7.2 Test Cases

Testing was done in the Developer Edition org using dedicated test records. Screenshots were captured for each major feature, and the complete working was recorded in the project demonstration video.

ID	Feature	Test Scenario	Expected Result	Result
TC-01	Objects & Relationships	Create Event with Client and Venue lookups	Event saved with related Client and Venue	Passed
TC-02	Validation Rule	Save a Client with an invalid email ("Validation Test Client")	Error: Please Enter Valid Email Address	Passed
TC-03	Lookup Filter	Create Feedback and open the Event lookup ("Lookup Filter Test")	Only events of the selected client are listed	Passed
TC-04	Venue Automation	Confirm an event with a venue ("Venue Automation Test")	Venue Availability Status becomes Reserved	Passed
TC-05	Double Booking	Create a second event at the same venue on the same date	Error: This Venue is already booked on this date.	Passed
TC-06	Reminder Flow	Confirm an event ("Reminder Flow Test")	Scheduled path queued; client emailed 3 days before event	Passed
TC-07	Approval Process	Set status Pending Cancellation ("Final Cancellation Test")	Request submitted, approved; status becomes Canceled	Passed
TC-08	Batch / Scheduled Apex	Run job for a past event ("Past Event Completion Test")	Event Status becomes Completed	Passed
TC-09	Reports & Dashboard	Run report and open dashboard	Chart shows upcoming events by month with budget total	Passed
TC-10	Security	Review profiles, roles, permission sets, and OWD	Access limited as configured per role	Passed
TC-11	Event Vendor Association	Create Event Vendor record	Vendor associated with event	Passed

ID	Feature	Test Scenario	Expected Result	Result
TC-12	Feedback Creation	Create Feedback with matching client/event	Record accepted, relationships correct	Passed
TC-13	Formula Field	Create Event with Wedding type	Budget automatically calculated as 50,000	Passed
TC-14	Venue Release	Cancel an approved event	Venue status becomes Available	Passed
TC-15	Email Notification	Submit cancellation request	Email alert sent to Event Coordinator role	Passed

Table 14: Test Cases

7.3 Testing Results

All features behaved as expected during testing. Automation executed correctly, approvals updated event status, venue validation prevented conflicting bookings, and dashboards reflected the loaded data.

Key Testing Outcomes:

Area	Test Result	Evidence
Validation	Invalid email addresses were rejected with appropriate error message	TC-02
Formula	Event budgets were automatically calculated based on event type	TC-13
Venue Automation	Confirmed events reserved the associated venue; cancelled events made the venue available	TC-04, TC-14
Double Booking	Duplicate venue booking was prevented with error message	TC-05
Batch Apex	Past events were successfully changed to Completed	TC-08
Approval	Cancellation requests were successfully routed to the configured manager	TC-07
Feedback	Feedback records were successfully associated with the correct clients and events	TC-03, TC-12
Reminder Flow	Scheduled path was queued for confirmed events	TC-06
Security	Access limited as configured per role	TC-10

Area	Test Result	Evidence
Reports	Chart shows upcoming events by month with budget total	TC-09

Table 15: Testing Results

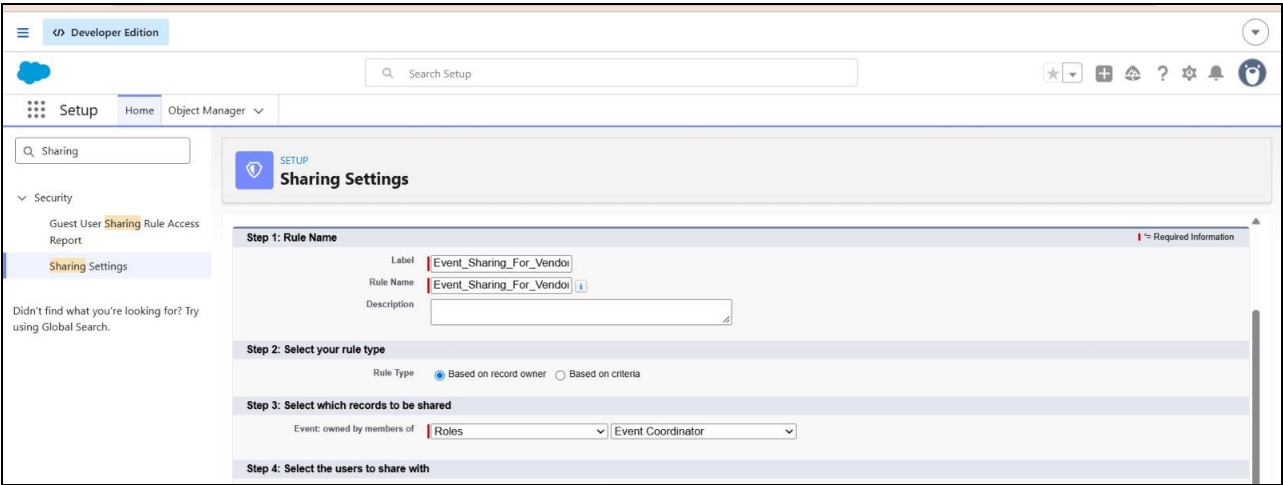


Figure 93: Past Event Completion Test Record

Performance Observations:

Operation	Response Time	Notes
Record Creation	< 1 second	Standard Salesforce performance
Validation Rule	< 1 second	Immediate feedback on save
Approval Submission	< 2 seconds	Includes email alert processing
Flow Execution	< 3 seconds	Scheduled path queued successfully
Batch Apex	< 10 seconds	For test data volume
Report Generation	< 5 seconds	For sample data volume
Dashboard Load	< 3 seconds	With single widget

Issues Found and Resolved:

Issue	Resolution
Initial formula syntax error	Corrected CASE statement syntax
Approval process not routing correctly	Verified manager assignment on user records
Flow not triggering	Adjusted entry conditions to include "created or updated"
Double booking trigger false positive	Excluded Canceled and Rejected statuses
Permission set not granting access	Verified object settings and assignments

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

The EventForce Management System provides several advantages:

1. Centralized Data Management

All event, client, vendor, venue, and feedback data is managed within Salesforce, providing a single source of truth for event operations. This eliminates data fragmentation and ensures consistency across all records.

2. Process Automation

Flows automate reminders and other business processes, significantly reducing manual effort and human error. The three-day reminder Flow ensures clients receive timely communication without manual follow-up.

3. Prevents Double Booking

Apex Trigger and VenueStatusHelper validate venue availability and help prevent booking conflicts, ensuring accurate venue scheduling. The PreventDoubleBooking trigger checks for existing events at the same venue on the same date before allowing a new booking.

4. Controlled Cancellation Process

Approval Process provides structured handling of event cancellation requests with manager-based routing and email notifications. This ensures that cancellations are reviewed and approved before status changes.

5. Improved Operational Visibility

Reports and Dashboards provide visibility into event schedules, vendor activities, and feedback for better decision-making. Management can monitor upcoming events and budget totals at a glance.

6. Secure Access

Profiles, Roles, Permission Sets, OWD, and Sharing Rules provide controlled access to records based on user responsibilities. Each user sees only the data relevant to their role.

7. Scalable Solution

Salesforce provides a platform that can support future growth and additional business requirements. New objects, fields, and automation can be added incrementally.

8. Better Data Quality

Validation Rules and pre-validation during data migration help maintain accurate and consistent data. Email validation ensures valid contact information.

9. Automated Communication

Email notifications and reminders ensure timely communication with clients and stakeholders. The system automatically sends reminders, approval notifications, and cancellation notices.

10. Reduced Manual Work

Automation reduces repetitive administrative tasks, allowing staff to focus on higher-value activities. The system handles reminders, status updates, and approvals automatically.

11. Audit Trail

Salesforce provides field history tracking and approval history, creating a complete audit trail of changes and decisions.

12. Mobile Access

Salesforce mobile app allows users to access event information and manage records from anywhere.

13. Integration Ready

Salesforce APIs enable integration with external systems for payments, calendars, and messaging.

8.2 Disadvantages

The current implementation has some limitations:

1. Developer Edition Limitation

The project was implemented in a Salesforce Developer Edition for academic purposes and was not connected to a production environment. Developer Edition has storage and API limits.

2. Production Deployment Not Performed

Actual migration to production was outside the scope of this implementation. Real deployments require sandbox environments and change management processes.

3. Deployment Requires Additional Tools

Real-world deployment may require sandboxes, Change Sets, Salesforce DX, or DevOps/CI-CD tools. These were not used in the current implementation.

4. Configuration Complexity

Managing Flows, Apex, Approval Processes, and security configurations requires appropriate Salesforce knowledge and expertise. Changes must be carefully tested before deployment.

5. Ongoing Maintenance Required

Flows, Apex jobs, approval processes, validation rules, and sharing settings need regular monitoring and troubleshooting. Scheduled jobs must be monitored for failures.

6. No Payment Integration

Payment processing is not currently included in the system. Integration with payment gateways would be required for online payments.

7. No External Calendar Integration

External calendar systems are not integrated for schedule synchronization. Events must be manually added to personal calendars.

8. No Customer Self-Service Portal

Clients cannot directly book or manage events through a self-service portal. All interactions go through Event Coordinators.

9. Limited Mobile Support

A dedicated mobile application is not provided (though Salesforce mobile app can be used). The mobile experience may not be optimized for all use cases.

10. No SMS/WhatsApp Notifications

The system sends email notifications but does not support SMS or WhatsApp messaging, which may be preferred by some clients.

11. Batch Apex Governor Limits

Batch Apex has governor limits on records processed per execution. For very large data volumes, multiple batches may be required.

12. Email Sending Limits

Salesforce has daily email sending limits that may affect high-volume notification scenarios.

9. CONCLUSION

The EventForce Management System successfully demonstrates how Salesforce can be used to streamline and automate event planning and management operations. The project brings Events, Clients, Vendors, Venues, Feedback, and EventVendor data into a centralized platform.

The implementation combines Salesforce Lightning, Flows, Validation Rules, Formula Fields, Approval Processes, Apex Triggers, Apex Classes, Batch Apex, Scheduled Apex, security controls, Reports, and Dashboards to address key business requirements.

Key Achievements:

Achievement	Description
Centralized CRM	All event-related data is managed in one Salesforce platform
Automated Budget Calculation	Event budgets are automatically calculated based on event type
Data Quality	Validation rules ensure accurate email addresses and lookup filters ensure correct relationships
Venue Management	Venue availability is automatically updated based on event status
Double-Booking Prevention	Apex trigger prevents conflicting venue reservations

Achievement	Description
Controlled Cancellation	Approval process ensures structured handling of cancellation requests
Automated Reminders	Three-day reminder Flow ensures timely client communication
Automatic Event Completion	Batch Apex marks past events as Completed
Security	Profiles, Roles, Permission Sets, OWD, and Sharing Rules control access
Operational Visibility	Reports and Dashboards provide management insights

Table 16: Key Achievements

The project also demonstrates the importance of data migration, testing, security, documentation, and maintenance in a Salesforce implementation.

Although the solution was developed and tested in a Developer Edition Org and actual production deployment was not performed, the project reflects real-world Salesforce implementation practices and provides a foundation for future scalability and enhancement.

Lessons Learned:

1. **Declarative First:** Using Flows and validation rules reduces development time and maintenance
2. **Security by Design:** Access controls should be planned from the beginning
3. **Test Early:** Testing each component as it is built prevents integration issues
4. **Document Everything:** Proper documentation is essential for maintenance and knowledge transfer
5. **Governor Limits:** Understanding Salesforce governor limits is critical for Apex development
6. **User Experience:** The interface should be intuitive for users with different technical backgrounds

The four-member team successfully collaborated to deliver a complete, tested, and documented Salesforce CRM solution that addresses the identified business problems of Event Planner Solutions.

10. FUTURE SCOPE

The EventForce Management System can be further enhanced to support larger-scale event operations and improve automation, integration, and user experience.

1. Production Deployment

Deploy the solution from development to a production Salesforce environment using appropriate deployment tools (sandboxes, Change Sets, Salesforce DX).

2. Advanced Event Automation

Extend automation for additional event planning and operational activities such as automated vendor assignments, resource allocation, and task management.

3. Enhanced Vendor Management

Introduce more advanced vendor tracking and performance management capabilities, including vendor ratings, feedback analysis, and contract management.

4. Advanced Analytics

Add more reports and dashboards for deeper insights into events, venues, vendors, and client feedback. Implement Salesforce CRM Analytics for advanced business intelligence.

5. External Integrations

Integrate Salesforce with external systems and services required for real-world event operations:

Integration	Purpose
Payment Gateways	Online payments for event bookings
Calendar Systems	Google Calendar, Microsoft Outlook synchronization
Messaging Platforms	WhatsApp, SMS notifications
Email Marketing	Campaign management and bulk communications
Accounting Systems	Invoice generation and financial tracking

6. Mobile Accessibility

Enhance the solution for convenient event management through mobile devices with a dedicated mobile application.

7. Customer Self-Service Portal

Implement a customer portal where clients can directly request events, view status, and provide feedback.

8. AI-Powered Features

Implement AI-based event recommendations, venue suggestions, vendor matching, and feedback summarization using Salesforce Einstein.

9. Scalability Enhancements

Expand the system to support larger numbers of events, clients, vendors, venues, and users.

10. Continuous Maintenance & Optimization

Regularly improve Flows, Apex, validation rules, sharing settings, and other configurations as business requirements evolve.

11. Multi-Language Support

Add support for multiple languages to serve clients from different regions.

12. Advanced Reporting

Implement custom report types and joined reports for more complex analytics.

13. Chatter Integration

Enable Chatter for collaboration between team members on event records.

14. Document Management

Integrate with Salesforce Files or external document management systems for contracts and event documents.

15. Audit and Compliance

Enhance audit capabilities with field history tracking and compliance reporting.

11. TEAM CONTRIBUTION

This project was developed as a four-member team project. The implementation was divided across multiple areas to ensure collaborative development.

Team Member 1: Salesforce Data Model & Application

Responsibilities:

- Project setup and Salesforce Developer account creation
- Salesforce application (Event Planner Lightning App) configuration
- Custom objects creation (Event, Client, Vendor, Venue, Feedback, EventVendor)
- Custom tabs creation for all objects
- Fields and relationships (Lookup, Master-Detail, Many-to-Many)
- Sample data creation and loading
- Object-level documentation

Deliverables:

- Six custom objects with all required fields
- Complete relationship model
- Lightning App with navigation
- Sample data for testing
- Object documentation

Team Member 2: Automation & Workflow

Responsibilities:

- Validation rules implementation (Email_Valid_Address)
- Formula fields (Event Budget calculation)
- Lookup filters (Feedback → Event filter)
- Approval Process (Event_Cancellation_Process)
- Email templates for notifications
- Client reminder Flow (3-Day Reminder)
- Automation testing

Deliverables:

- Email validation rule
- Event Budget formula field

- Lookup filter on Feedback object
- Complete approval process with email alerts
- Record-triggered Flow with scheduled path

Team Member 3: Apex Development

Responsibilities:

- VenueStatusHelper Apex class
- EventVenueStatusTrigger
- PreventDoubleBooking Apex trigger
- BatchCompleteEvents Batch Apex class
- ScheduleCompleteEvents Schedulable Apex class
- Apex testing and debugging
- Scheduled job configuration

Deliverables:

- Venue status automation logic
- Double-booking prevention trigger
- Batch processing for event completion
- Scheduled job for daily execution
- Apex test coverage

Team Member 4: Security, Reporting & Testing

Responsibilities:

- Profiles creation (Event Admin, Event Coordinator, Vendor Manager, Client)
- Roles hierarchy configuration
- User creation and assignment
- Permission Sets (Feedback Manager)
- Organization-Wide Defaults configuration
- Sharing Rules (Event_Sharing_For_Vendors)
- Reports (Upcoming Events by Month)
- Dashboard (EventForce Operations Dashboard)
- Functional testing and documentation

Deliverables:

- Complete security model
- Reports and dashboard
- Test cases and results
- Project documentation

Collaborative Activities

All team members participated in:

Activity	Description
Requirement Analysis	Joint sessions to identify and document requirements
Sprint Planning	Collaborative estimation and task assignment
Code Reviews	Peer review of Apex code and Flows
Integration Testing	End-to-end testing of all components
Demo Preparation	Script preparation and rehearsal
Documentation	Joint review and finalization

Team Contribution Matrix:

Area	Member 1	Member 2	Member 3	Member 4
Data Model	●●●	●	●	●
Objects & Fields	●●●	●	●	●
Relationships	●●●	●	●	●
Validation Rules	●	●●●	●	●
Formula Fields	●	●●●	●	●
Flows	●	●●●	●	●
Approval Process	●	●●●	●	●
Apex Classes	●	●	●●●	●
Apex Triggers	●	●	●●●	●
Batch Apex	●	●	●●●	●
Scheduled Apex	●	●	●●●	●

Area	Member 1	Member 2	Member 3	Member 4
Profiles	•	•		